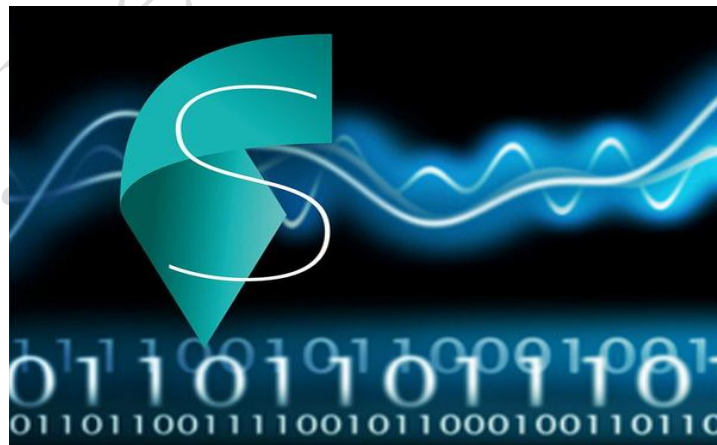




CYBER SECURITY TECHNOLOGY ENGINEERING DEPARTMENT

DIGITAL SIGNAL PROCESSING THIRD STAGE

Lect.1 Signals, Systems, and Digital Signal Processing



Asst. Lect. Haider Saad

2025 - 2026

1. Introduction

This lecture introduces the fundamental concepts of signals, systems, and digital signal processing (DSP).

2. Signals, Systems, and Signal Processing

A signal is a function that conveys information about the behavior or attributes of some phenomenon. In engineering, a signal is defined as a physical quantity that varies with one or more independent variables such as time, space, or frequency. Mathematically, a signal can be expressed as a function $s(t)$, where t represents time. These signals may be described mathematically, such as:

- $s_1(t) = 5t \rightarrow$ A signal that grows linearly with time.
- $s_2(t) = 20t^2 \rightarrow$ A signal with quadratic growth.
- $S(x, y) = 3x + 2xy + 10y^2 \rightarrow$ A two-dimensional signal.

$$\begin{aligned} \blacksquare s_1(t) &= 5t & \text{or} & & s_1(t) &= 5t^2 & \rightarrow \text{one variable} \\ \blacksquare S(x, y) &= 3x + 4xy + 6x^2 & & & & & \rightarrow \text{two variables } x \text{ and } y \end{aligned}$$

In real-world applications, many signals are not easily expressed by simple functions. For example, a speech signal is highly complex and cannot be accurately described by basic equations. Instead, it can be approximated as a combination of sinusoidal components with varying amplitudes and frequencies.

Figure 1.1.1 shows an example of a speech signal:

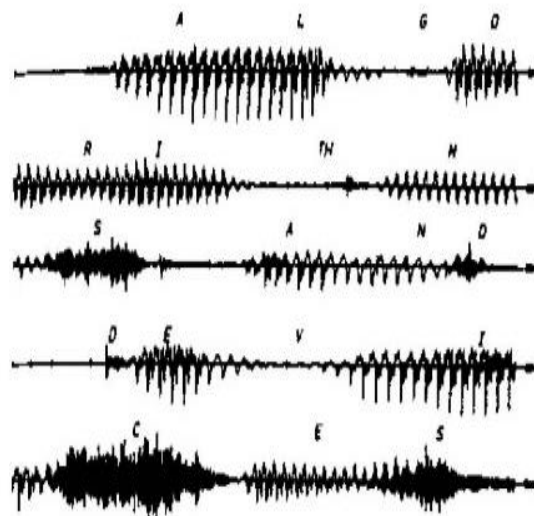
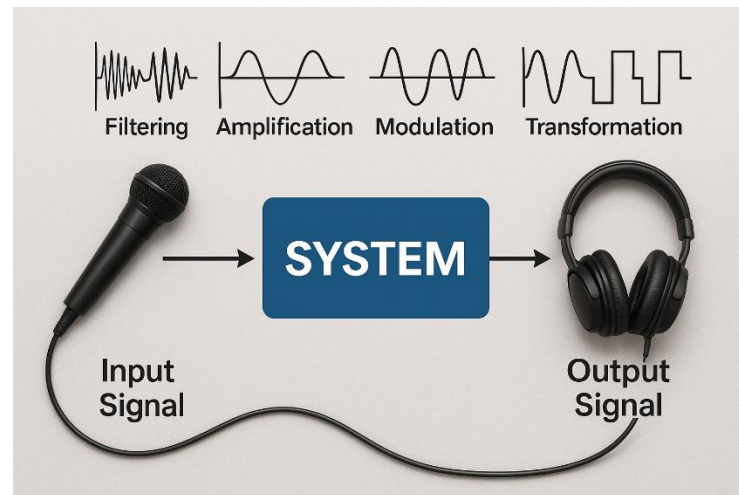


Figure 1.1.1
Example of a speech signal.

3. What is a System?

A system is any process that produces an output signal in response to an input signal. In signal processing, systems perform operations such as filtering, amplification, modulation, or transformation of signals.

For example, an audio equalizer is a system that filters frequency components of an audio signal.



4. Signal Processing

Signal processing refers to the analysis, manipulation, and interpretation of signals. There are **two main types of signal processing: analog and digital**. In analog signal processing, operations are carried out using continuous hardware components. In digital signal processing (DSP), operations are performed using algorithms executed on digital hardware such as microprocessors or digital signal processors.

5. Basic Elements of a Digital Signal Processing System

Most natural signals are analog and must be converted to digital form before digital processing. This is achieved through an Analog-to-Digital (A/D) converter, followed by a digital signal processor, and optionally a Digital-to-Analog (D/A) converter.

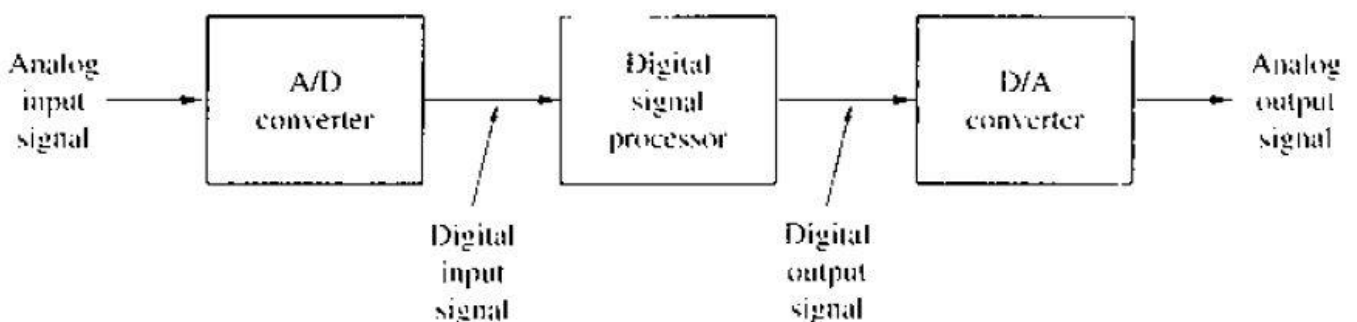


Figure 1.1.3 Block diagram of a digital signal processing system.

4. Advantages of Digital Signal Processing

Digital signal processing offers several advantages over analog processing:

- Flexibility – Easily reprogrammed or updated.
- High precision – Better control over numeric computations.
- Storage – Easy to store, retrieve, and transmit.
- Reliability – Less affected by noise, and environmental conditions.
- Integration – DSP can be implemented on compact, low-power chips

5. Classification of Signals

Signals can be classified in multiple ways:

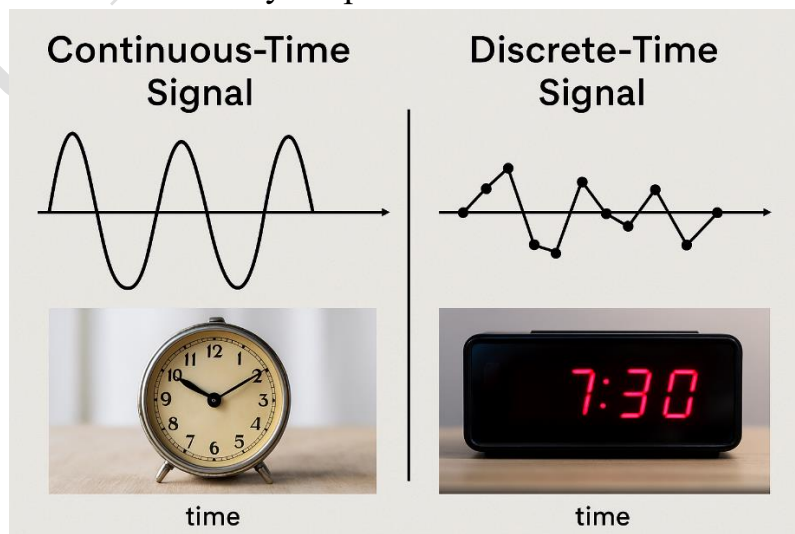
a) Based on the number of variables:

- One-dimensional (e.g., audio).
- Multidimensional (e.g., images, video).



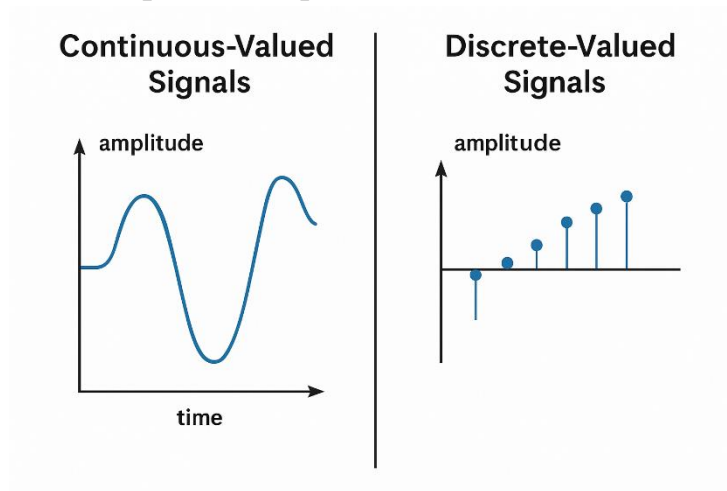
b) Based on time definition:

- Continuous-time signals – Defined for all time (analog).
- Discrete-time signals – Defined only at specific instances.



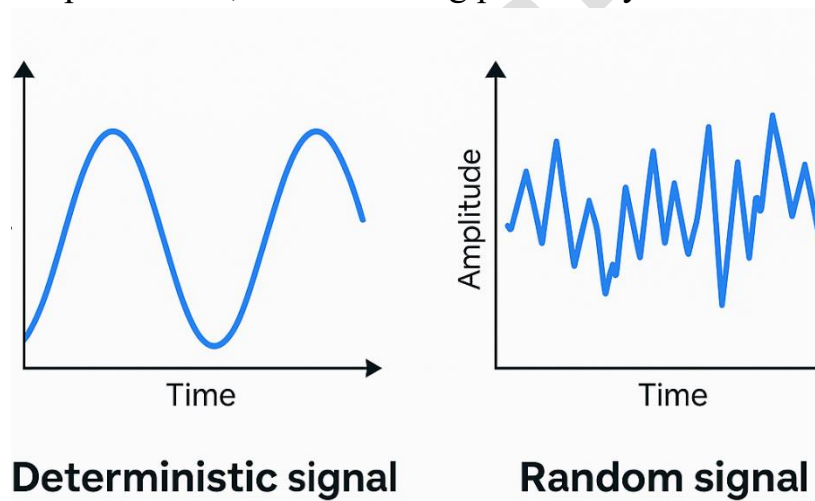
c) Based on amplitude range:

- Continuous-valued signals – Amplitude can take any value.
- Discrete-valued signals – Amplitude is quantized.



d) Based on predictability:

- Deterministic signals – Predictable and modeled by equations.
- Random signals – Unpredictable, modeled using probability.

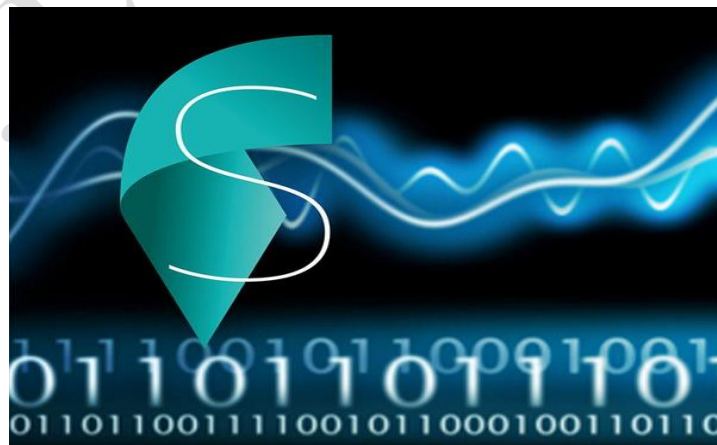




CYBER SECURITY TECHNOLOGY ENGINEERING DEPARTMENT

DIGITAL SIGNAL PROCESSING THIRD STAGE

Lect.2 The Concept of Frequency in Continuous and Discrete Time Signal



Asst. Lect. Haider Saad

2025 - 2026

1. Introduction

Frequency represents the number of cycles a signal completes per unit of time. It plays a central role in understanding signal behavior in both continuous and discrete domains. In continuous-time, frequency is measured in Hz (cycles/second), while in discrete-time, it is measured in cycles per sample. Frequency is crucial in analyzing sinusoidal and periodic signals.

2. Continuous-Time Sinusoidal Signals

A continuous-time sinusoidal signal is represented by:

$$x_a(t) = A \cos(\Omega t + \theta) \quad (\text{Eq. 1.3.1})$$

$$x_a(t) = A \cos(\Omega t + \theta), \quad -\infty < t < \infty \quad (1.3.1)$$

where:

- A is the amplitude of the sinusoid,
- Ω is the angular frequency (radians/sec),
- θ is the phase (radians).
- x_a represent analog signal.

The angular frequency is related to the frequency in Hz (F) by:

$$\Omega = 2\pi F \quad (\text{Eq. 1.3.2})$$

$$\Omega = 2\pi F \quad (1.3.2)$$

Thus:

$$x_a(t) = A \cos(2\pi F t + \theta) \quad (\text{Eq. 1.3.3})$$

$$x_a(t) = A \cos(2\pi F t + \theta), \quad -\infty < t < \infty \quad (1.3.3)$$

We will use both forms, (1.3.1) and (1.3.3), in representing sinusoidal signals.

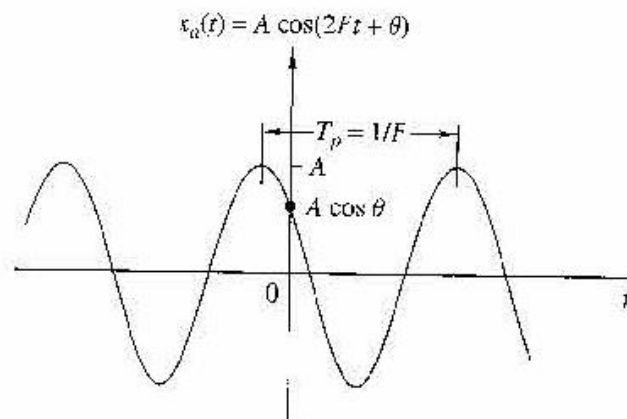


Figure 1.3.1
Example of an analog
sinusoidal signal.

2.1 Properties of Continuous-Time Sinusoids

- A1.** For every fixed value of the frequency F , $x_a(t)$ is periodic. Indeed, it can easily be shown, using elementary trigonometry, that

$$x_a(t + T_p) = x_a(t)$$

where $T_p = 1/F$ is the fundamental period of the sinusoidal signal.

- A2.** Continuous-time sinusoidal signals with distinct (different) frequencies are themselves distinct
- A3.** Increasing the frequency F results in an increase in the rate of oscillation of the signal, in the sense that more periods are included in a given time interval.

We observe that for $F = 0$, the value $T_p = \infty$ is consistent with the fundamental relation $F = 1/T_p$. Due to continuity of the time variable t , we can increase the frequency F , without limit, with a corresponding increase in the rate of oscillation.

The relationships we have described for sinusoidal signals carry over to the class of complex exponential signals

$$x_a(t) = Ae^{j(\Omega t + \theta)} \quad (1.3.4)$$

This can easily be seen by expressing these signals in terms of sinusoids using the Euler identity

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi \quad (1.3.5)$$

By definition, frequency is an inherently positive physical quantity. This is obvious if we interpret frequency as the number of cycles per unit time in a periodic signal. However, in many cases, only for mathematical convenience, we need to introduce negative frequencies. To see this we recall that the sinusoidal signal (1.3.1) may be expressed as

$$x_a(t) = A \cos(\Omega t + \theta) = \frac{A}{2} e^{j(\Omega t + \theta)} + \frac{A}{2} e^{-j(\Omega t + \theta)} \quad (1.3.6)$$

- Reason for using this representation: Math becomes much simpler when dealing with exponents instead of sine/cosine (especially in transformations and spectral analysis).

3. Discrete-Time Sinusoidal Signals

A discrete-time sinusoid is given by:

$$x(n) = A \cos(\omega n + \theta) \quad (\text{Eq. 1.3.7})$$

ω is the frequency in radians/sample. If we use **normalized frequency** f :

$$\omega = 2\pi f \quad (\text{Eq. 1.3.8}), \text{ so}$$

$$x(n) = A \cos(2\pi f n + \theta) \quad (\text{Eq. 1.3.9})$$

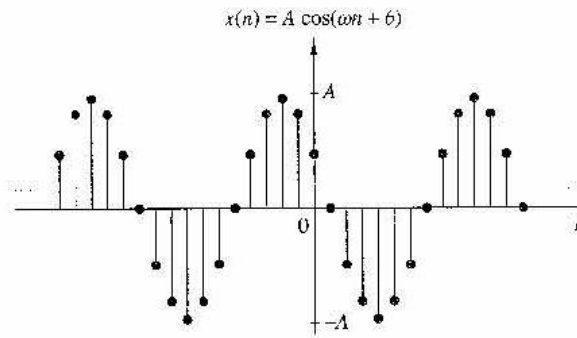


Figure 1.3.3
Example of a discrete-time sinusoidal signal ($\omega = \pi/6$ and $\theta = \pi/3$).

3.1 Properties of Discrete-Time Sinusoids

In contrast to continuous-time sinusoids, the discrete-time sinusoids are characterized by the following properties:

B1. A discrete-time sinusoid is periodic only if its frequency f is a rational number.

By definition, a discrete-time signal $x(n)$ is periodic with period N ($N > 0$) if and only if

$$x(n + N) = x(n) \quad \text{for all } n \quad (1.3.10)$$

The smallest value of N for which (1.3.10) is true is called the *fundamental period*.

The proof of the periodicity property is simple. For a sinusoid with frequency f_0 to be periodic, we should have

$$\cos[2\pi f_0(N + n) + \theta] = \cos(2\pi f_0 n + \theta)$$

This relation is true if and only if there exists an integer k such that

$$2\pi f_0 N = 2k\pi$$

or, equivalently,

$$f_0 = \frac{k}{N} \quad (1.3.11)$$

According to (1.3.11), a discrete-time sinusoidal signal is periodic only if its frequency f_0 can be expressed as the ratio of two integers (i.e., f_0 is rational).

To determine the fundamental period N of a periodic sinusoid, we express its frequency f_0 as in (1.3.11) and cancel common factors so that k and N are relatively prime. Then the fundamental period of the sinusoid is equal to N . Observe that a small change in frequency can result in a large change in the period. For example, note that $f_1 = 31/60$ implies that $N_1 = 60$, whereas $f_2 = 30/60$ results in $N_2 = 2$.

B2. Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical.

To prove this assertion, let us consider the sinusoid $\cos(\omega_0 n + \theta)$. It easily follows that

$$\cos[(\omega_0 + 2\pi)n + \theta] = \cos(\omega_0 n + 2\pi n + \theta) = \cos(\omega_0 n + \theta) \quad (1.3.12)$$

As a result, all sinusoidal sequences

$$x_k(n) = A \cos(\omega_k n + \theta), \quad k = 0, 1, 2, \dots \quad (1.3.13)$$

where

$$\omega_k = \omega_0 + 2k\pi, \quad -\pi \leq \omega_0 \leq \pi$$

are *indistinguishable* (i.e., *identical*). Any sequence resulting from a sinusoid with a frequency $|\omega| > \pi$, or $|f| > \frac{1}{2}$, is identical to a sequence obtained from a sinusoidal signal with frequency $|\omega| < \pi$. Because of this similarity, we call the sinusoid having the frequency $|\omega| > \pi$ an *alias* of a corresponding sinusoid with frequency $|\omega| < \pi$. Thus we regard frequencies in the range $-\pi \leq \omega \leq \pi$, or $-\frac{1}{2} \leq f \leq \frac{1}{2}$, as unique

and all frequencies $|\omega| > \pi$, or $|f| > \frac{1}{2}$, as aliases. The reader should notice the difference between discrete-time sinusoids and continuous-time sinusoids, where the latter result in distinct signals for Ω or F in the entire range $-\infty < \Omega < \infty$ or $-\infty < F < \infty$.

B3. The highest rate of oscillation in a discrete-time sinusoid is attained when $\omega = \pi$ (or $\omega = -\pi$) or, equivalently, $f = \frac{1}{2}$ (or $f = -\frac{1}{2}$).

To illustrate this property, let us investigate the characteristics of the sinusoidal signal sequence

$$x(n) = \cos \omega_0 n$$

when the frequency varies from 0 to π . To simplify the argument, we take values of $\omega_0 = 0, \pi/8, \pi/4, \pi/2, \pi$ corresponding to $f = 0, \frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}$, which result in periodic sequences having periods $N = \infty, 16, 8, 4, 2$, as depicted in Fig. 1.3.4. We note that the period of the sinusoid decreases as the frequency increases. In fact, we can see that the rate of oscillation increases as the frequency increases.

To see what happens for $\pi \leq \omega_0 \leq 2\pi$, we consider the sinusoids with frequencies $\omega_1 = \omega_0$ and $\omega_2 = 2\pi - \omega_0$. Note that as ω_1 varies from π to 2π , ω_2 varies from π to 0. It can be easily seen that

$$\begin{aligned} x_1(n) &= A \cos \omega_1 n = A \cos \omega_0 n \\ x_2(n) &= A \cos \omega_2 n = A \cos(2\pi - \omega_0)n \\ &= A \cos(-\omega_0 n) = x_1(n) \end{aligned} \quad (1.3.14)$$

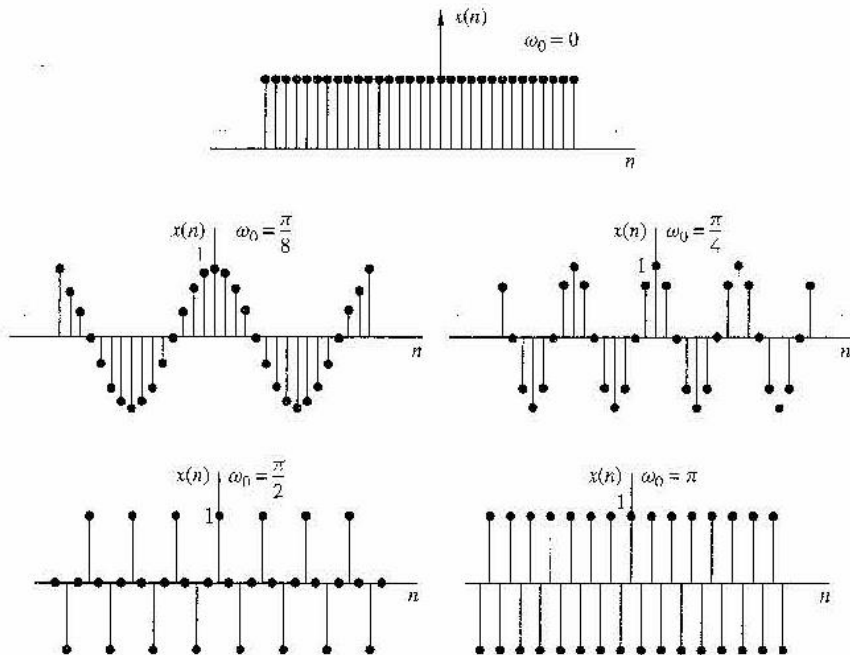


Figure 1.3.4 Signal $x(n) = \cos \omega_0 n$ for various values of the frequency ω_0 .

Hence ω_2 is an alias of ω_1 . If we had used a sine function instead of a cosine function, the result would basically be the same, except for a 180° phase difference between the sinusoids $x_1(n)$ and $x_2(n)$. In any case, as we increase the relative frequency ω_0 of a discrete-time sinusoid from π to 2π , its rate of oscillation decreases. For $\omega_0 = 2\pi$ the result is a constant signal, as in the case for $\omega_0 = 0$. Obviously, for $\omega_0 = \pi$ (or $f = \frac{1}{2}$) we have the highest rate of oscillation.

As for the case of continuous-time signals, negative frequencies can be introduced as well for discrete-time signals. For this purpose we use the identity

$$x(n) = A \cos(\omega n + \theta) = \frac{A}{2} e^{j(\omega n + \theta)} + \frac{A}{2} e^{-j(\omega n + \theta)} \quad (1.3.15)$$

4. Harmonically Related Complex Exponentials

These are complex exponentials with frequencies that are integer multiples of a fundamental frequency F_0 .

Continuous-time: $S_k(t) = e^{(j2\pi k F_0 t)}$ (Eq. 1.3.16)

Discrete-time: $S_k(n) = e^{(j2\pi kn/N)}$ (Eq. 1.3.18)

A periodic signal can be written as:

$$x_a(t) = \sum c_k * S_k(t) \quad (\text{Eq. 1.3.17})$$

$$x(n) = \sum c_k * S_k(n) \quad (\text{Eq. 1.3.20})$$

EXAMPLE 1.3.1

Stored in the memory of a digital signal processor is one cycle of the sinusoidal signal

$$x(n) = \sin\left(\frac{2\pi n}{N} + \theta\right)$$

where $\theta = 2\pi q/N$, where q and N are integers

- Determine how this table of values can be used to obtain values of harmonically related sinusoids having the same phase
- Determine how this table can be used to obtain sinusoids of the same frequency but different phase.

Solution.

- Let $x_k(n)$ denote the sinusoidal signal sequence

$$x_k(n) = \sin\left(\frac{2\pi nk}{N} + \theta\right)$$

This is a sinusoid with frequency $f_k = k/N$, which is harmonically related to $x(n)$. But $x_k(n)$ may be expressed as

$$\begin{aligned} x_k(n) &= \sin\left[\frac{2\pi(kn)}{N} + \theta\right] \\ &= x(kn) \end{aligned}$$

Thus we observe that $x_k(0) = x(0)$, $x_k(1) = x(k)$, $x_k(2) = x(2k)$, and so on. Hence the sinusoidal sequence $x_k(n)$ can be obtained from the table of values of $x(n)$ by taking every k th value of $x(n)$, beginning with $x(0)$. In this manner we can generate the values of all harmonically related sinusoids with frequencies $f_k = k/N$ for $k = 0, 1, \dots, N-1$.

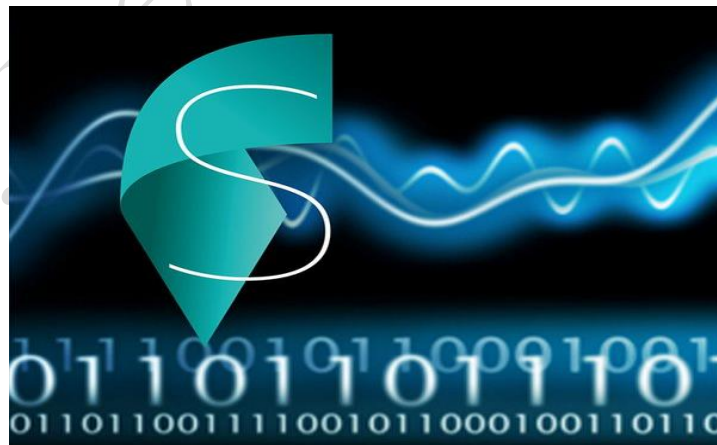
- We can control the phase θ of the sinusoid with frequency $f_k = k/N$ by taking the first value of the sequence from memory location $q = \theta N/2\pi$, where q is an integer. Thus the initial phase θ controls the starting location in the table and we wrap around the table each time the index (kn) exceeds N .



CYBER SECURITY TECHNOLOGY ENGINEERING DEPARTMENT

DIGITAL SIGNAL PROCESSING THIRD STAGE

Lect.3,4 Analog-to-Digital and Digital-to-Analog Conversions



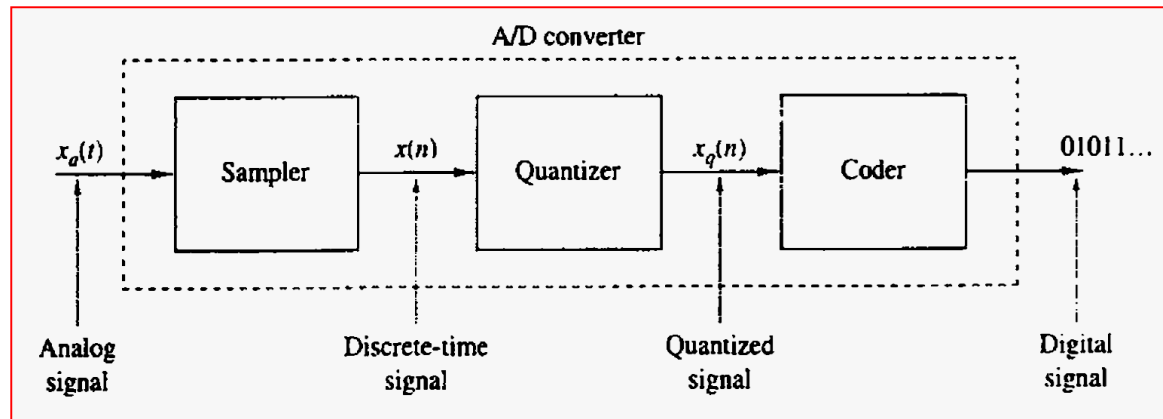
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Analog – to – digital and Digital – to Analog Conversion:

Most signals of practical interest, such as biological signals are analog. To process analog signals by digital means, it is first necessary to convert them in digital form. That is, to convert them to a sequence of numbers having finite precision. This procedure is called analog – to – digital (A/D) conversion.

Conceptually, we view A/D conversion as a three – step process. This process is illustrated in the figure below

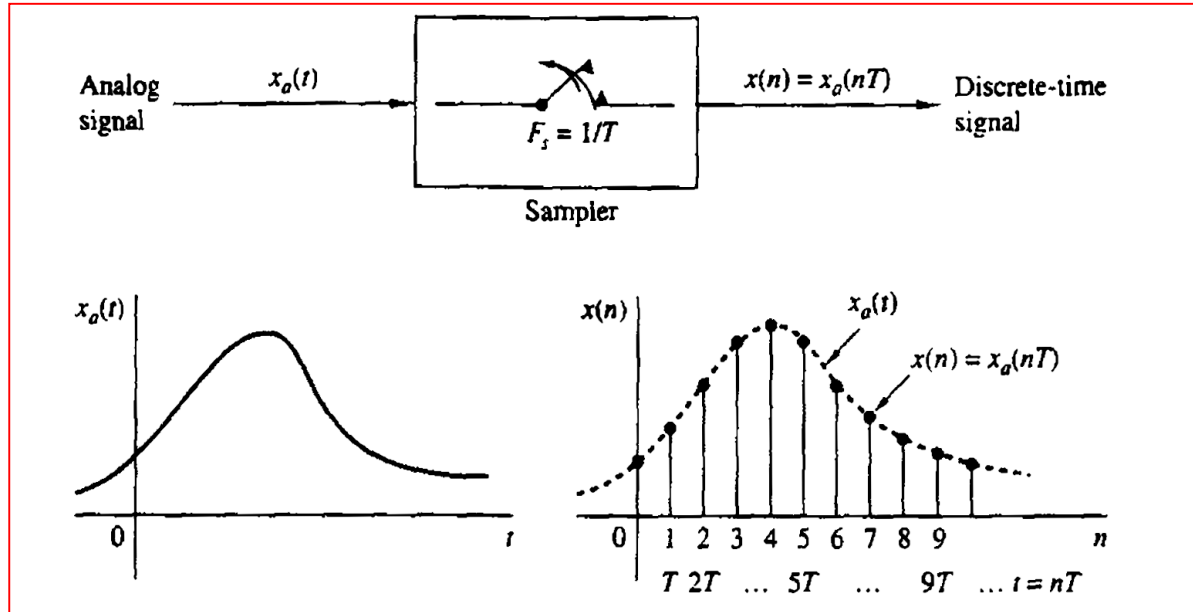


1. **Sampling.** This is the conversion of continuous – time signal into a discrete – time signal obtained by taking “samples” of the continuous – time signal at discrete – time instants. Thus, if $x_a(t)$ is the input to the sampler, the output is $x_a(nT) \equiv x(n)$, where T is called the sampling interval.
2. **Quantization.** This is the conversion of a discrete – time continuous – valued signal into a discrete – time, discrete – valued (digital) signal. The value of each signal sample is represented by a value selected from a finite set of possible values. The difference between the unquantized sample $x(n)$ and the quantized output $x_q(n)$ is called the quantization error.
That is, we round each sample to the nearest value from a pre-defined set of levels (e.g., 256 levels if the encoding is 8 bits)
3. **Coding.** In the coding process, each discrete value $x_q(n)$ is represented by a b – bit binary sequence.

Sampling of Analog Signals.

There are many ways to sample an analog signal. We limit our discussion to periodic or uniform sampling, which is the type of sampling used most often in practice. This is described by the relation:

$$x(n) = x_a(nT), \quad -\infty < n < \infty$$



The time interval T between successive samples is called the sampling period or sample interval and its reciprocal $1/T = F_s$ is called the sampling rate or sampling frequency.

The variables t and n are linearly related through the sampling period T or, equivalently, through the sampling rate $F_s = 1/T$, as

$$t = nT = \frac{n}{F_s}$$

As a consequence of above equation, there exists a relationship between the frequency variable F (or Ω) for analog signals and the frequency variable f (or ω) for discrete-time signals. To establish this relationship, consider an analog sinusoidal signal of the form:

$$\begin{aligned} x_a(t) &= A \cos(2\pi Ft + \theta) \\ x_a(nT) \equiv x(n) &= A \cos(2\pi FnT + \theta) \\ &= A \cos\left(\frac{2\pi nF}{F_s} + \theta\right) \end{aligned}$$

The frequency variables F and f are linearly related as:

$$f = \frac{F}{F_s}$$

Or, equivalently, as $\omega = \Omega T$

The frequency variable f is relative or normalized frequency. We can use f to determine the frequency F in hertz only if the sampling frequency F_s is known.

The relations are summarized in following table

Continuous – time signals		Discrete – time signals	
$\Omega = 2\pi F$		$\omega = 2\pi f$	
$\frac{\text{radians}}{\text{sec}}$	Hz	$\frac{\text{radians}}{\text{sample}}$	$\frac{\text{cycles}}{\text{sample}}$
		$\omega = \Omega T, f = F/F_s$	
		$\Omega = \omega/T, F = f.F_s$	
$-\infty < \Omega < \infty$		$-\pi/T \leq \omega \leq \pi/T$	
$-\infty < F < \infty$		$-F_s/2 \leq f \leq F_s/2$	

From these relations we observe that

$$f_{\max} = \frac{F_s}{2} = \frac{1}{2T}$$

$$\omega_{\max} = \pi F_s = \frac{\pi}{T}$$

The sampling theorem

If the highest frequency contained in an analog signal $x_a(t)$ is $F_{\max} = B$ and the signal is sampled at a rate $F_s > 2F_{\max} \equiv 2B$, then $x_a(t)$ can be exactly recovered from its sample values using the interpolation function:

$$g(t) = \frac{\sin 2\pi B t}{2\pi B t}$$

Thus $x_a(t)$ may be expressed as

$$x_a(t) = \sum_{n=-\infty}^{\infty} x_a\left(\frac{n}{F_s}\right) g\left(t - \frac{n}{F_s}\right)$$

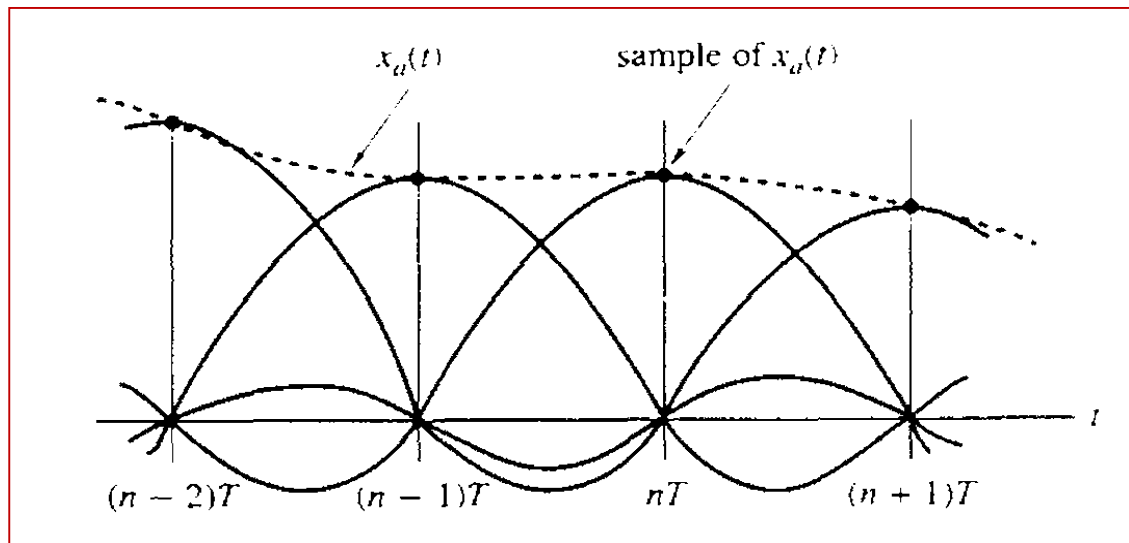
Where $x_a(n/F_s) = x_a(nT) \equiv x(n)$ are the samples of $x_a(t)$.

When the sampling of $x_a(t)$ is performed at the minimum sampling rate $F_s = 2B$, the reconstruction formula becomes

$$x_a(t) = \sum_{n=-\infty}^{\infty} x_a\left(\frac{n}{2B}\right) \frac{\sin 2\pi B(t - \frac{n}{2B})}{2\pi B(t - \frac{n}{2B})}$$

And the sampling rate $F_N = 2B$ is called Nyquist rate.

Figure below illustrate the ideal D/A conversion process using the interpolation function.



Example

Consider the analog signal

$$x_a(t) = 3\cos 50\pi t + 10\sin 300\pi t - \cos 100\pi t$$

What is the Nyquist rate for this signal?

Solution

The frequencies present in the signal above are

$$F_1 = 25 \text{ Hz} \quad F_2 = 150 \text{ Hz} \quad F_3 = 50 \text{ Hz}$$

$$\text{Thus } F_{\max} = 150 \text{ Hz and } F_s > 2F_{\max} = 300 \text{ Hz} = F_N$$

Example

Consider the analog signal

$$x_a = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12000\pi t$$

- What is the Nyquist rate for this signal?
- Assume now that we sample this signal using a sampling rate $F_s = 5000$ samples/sec. what is the discrete – time signal obtained after sampling?
- What is the analog signal $y_a(t)$ we can reconstruct from the samples if we use ideal interpolation?

Solution

- The frequencies existing in the analog signal are
 $F_1 = 1 \text{ KHz}, \quad F_2 = 3 \text{ KHz}, \quad F_3 = 6 \text{ KHz}$
 Thus $F_{\max} = 6 \text{ KHz}$, and according to the sampling theorem

$F_s > 2 F_{\max} = 12 \text{ KHz}$ and the Nyquist rate is $F_N = 12 \text{ KHz}$

b) Since we have chosen $F_s = 5 \text{ KHz}$, the folding frequency is

$$\frac{F_s}{2} = 2.5 \text{ KHz}$$

And this is the maximum frequency that can be represented uniquely by the sampled signal. We obtain

$$\begin{aligned} x(n) &= x_a(nT) = x_a\left(\frac{n}{F_s}\right) \\ &= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(\frac{3}{5}\right)n + 10\cos 2\pi\left(\frac{6}{5}\right)n \\ &= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(1 - \frac{2}{5}\right)n + 10\cos 2\pi\left(1 + \frac{1}{5}\right)n \\ &= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(-\frac{2}{5}\right)n + 10\cos 2\pi\left(\frac{1}{5}\right)n \end{aligned}$$

Now, using the periodicity of discrete-time frequency:

$$\cos(2\pi(f + k)n) = \cos(2\pi fn)$$

$$\sin(2\pi(f + k)n) = \sin(2\pi fn)$$

Finally, we obtain

$$x(n) = 13\cos 2\pi\left(\frac{1}{5}\right)n - 5\sin 2\pi\left(\frac{2}{5}\right)n$$

c) Since only the frequency components at 1 KHz and 2 KHz are present in the sampled signal, the analog signal we can recover is

$$y_a(t) = 13\cos 2000\pi t - 5\sin 4000\pi t$$

Which is obviously different from the original signal $x_a(t)$. This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.

Quantization of Continuous – Amplitude signal

The process of converting a discrete – time continuous – amplitude signal into a digital signal by expressing each sample value as a finite (instead of an infinite) number of digits, is called quantization. The error introduced in representing the continuous – valued signal by a finite set of discrete value levels is called quantization error or quantization noise.

We denote the quantizer operation on the samples $x(n)$ as $Q[x(n)]$ and let $x_q(n)$ denote the sequence of quantized samples at the output of the quantizer. Hence

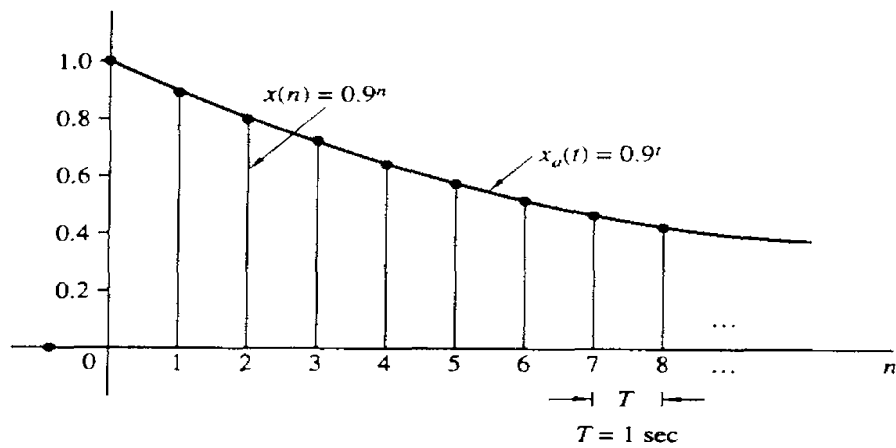
$$x_q(n) = Q[x(n)]$$

Then the quantization error is a sequence $e_q(n)$ defined as the difference between the quantized value and the actual sample value. Thus

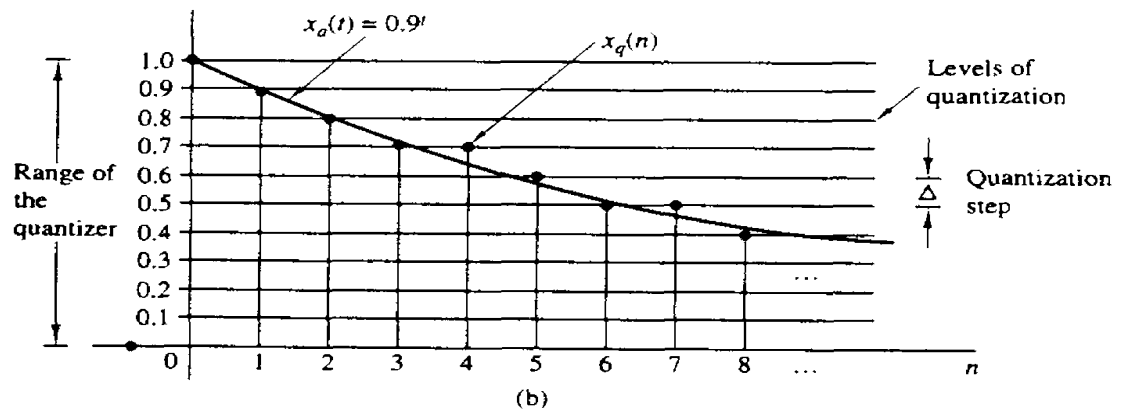
$$e_q(n) = x_q(n) - x(n)$$

Let us consider the discrete – time signal

$$x(n) = \begin{cases} 0.9^n, & n \geq 0 \\ 0 & n < 0 \end{cases}$$



(a)



(b)

TABLE 1.2 NUMERICAL ILLUSTRATION OF QUANTIZATION WITH ONE SIGNIFICANT DIGIT USING TRUNCATION OR ROUNDING

n	$x(n)$ Discrete-time signal	$x_q(n)$ (Truncation)	$x_q(n)$ (Rounding)	$e_q(n) = x_q(n) - x(n)$ (Rounding)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	-0.01
3	0.729	0.7	0.7	-0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	-0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	-0.03046721
9	0.387420489	0.3	0.4	0.012579511

Post test:-

Circle the correct answer:-

1- Sampling Process is:

- a- Is the process of converting continuous time signal to discrete time signal.
- b- Is the process of converting continuous time signal to digital signal.
- c- Is the process of converting discrete time signal to digital signal.
- d- Is the process of converting digital signal to analog signal.

2- Quantization Process is:

- a- Conversion of discrete time discrete valued signal to digital signal.
- b- Conversion of discrete time continuous valued signal to discrete time discrete valued signal.
- c- Conversion of discrete time continuous valued signal to continuous time discrete valued signal.
- d- Conversion of digital signal to continuous time discrete valued signal.

3- Coding Process is:

- a- Representing of continuous valued signal by binary numbers.
- b- Representing of continuous valued signal by real numbers.
- c- Representing of discrete valued signal by binary numbers.

key answer :-

1- Pre test:-

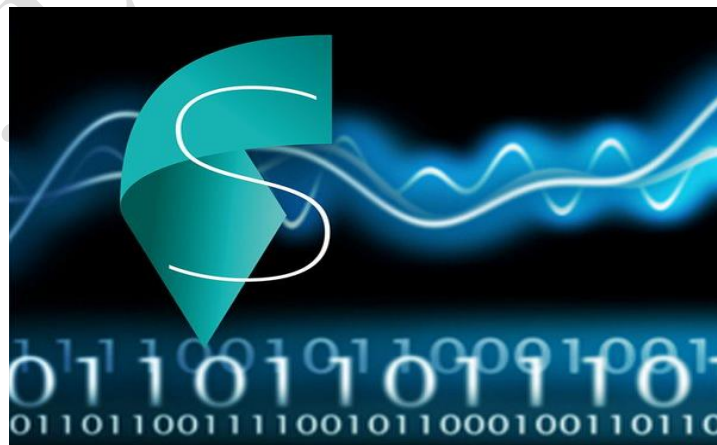
- 1. a**
- 2. b**
- 3. c**



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DIGITAL SIGNAL PROCESSING THIRD STAGE

Lect.5 Discrete Time Systems



Asst. Lect. Haider Saad

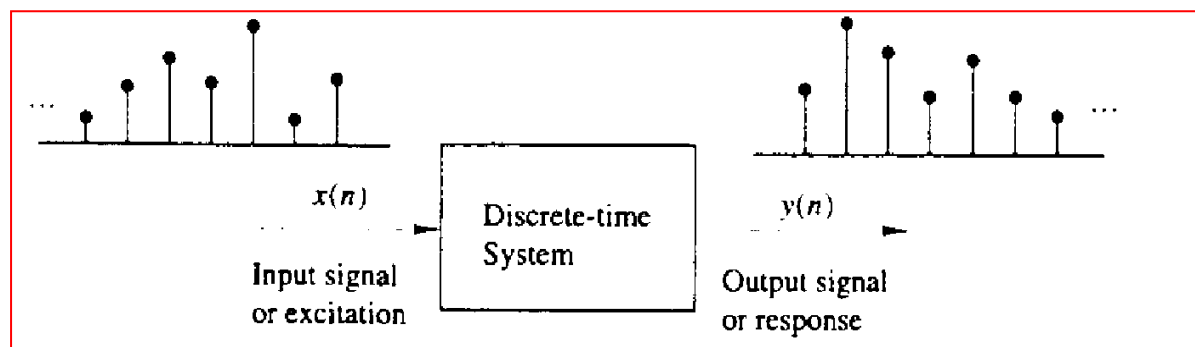
2025 - 2026

Discrete – Time Systems

In many applications of digital signal processing we wish to design a device or an algorithm that performs some prescribed operation on discrete – time signal. Such a device or algorithm is called a discrete – time system. More specifically, a discrete – time system is a device or algorithm that operates on a discrete – time signal called the input or excitation, according to some well – defined rule, to produce another discrete – time signal called the output or response of the system.

Input – output Description of Systems

The input – output description of a discrete – time system consists of a mathematical expression or a rule, which explicitly defines the relation between the input and output signals.



Block diagram representation of a discrete – time system

Example

Determine the response of the following systems to the input signal

$$x(n) = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

- a) $y(n) = x(n)$
- b) $y(n) = x(n - 1)$
- c) $y(n) = x(n + 1)$

Solution First, we determine explicitly the sample values of the input signal

$$x(n) = \{ \dots \dots 0, 3, 2, 1, 0, 1, 2, 3, 0 \dots \dots \}$$

Next, we determine the output of each system using its input – output relationship.

- a) In this case the output is exactly the same as the input signal. Such a system is known as the identity system.
- b) This system simply delays the input by one sample

$x(n)$	3	2	1	0	1	2	3
$x(n - 1)$	0	3	2	1	0	1	2

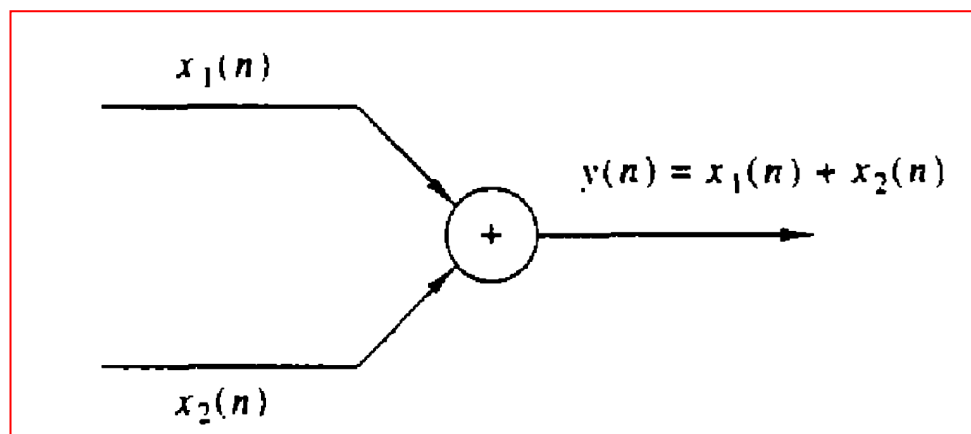
- c) In this case the system “advances” the input one sample into the future.

$x(n)$	3	2	1	0	1	2	3
$x(n + 1)$	2	1	0	1	2	3	0

Block Diagram representation of Discrete – Time Systems

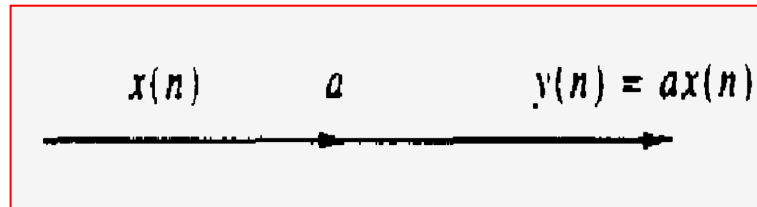
It is useful at this point to introduce a block diagram representation of discrete – time systems. For this purpose we need to define some basic building blocks that can be interconnected to form complex systems.

An adder. The figure below illustrate a system (adder) that performs the addition of two signal sequences to form another (the sum) sequence, which we denote as $y(n)$.

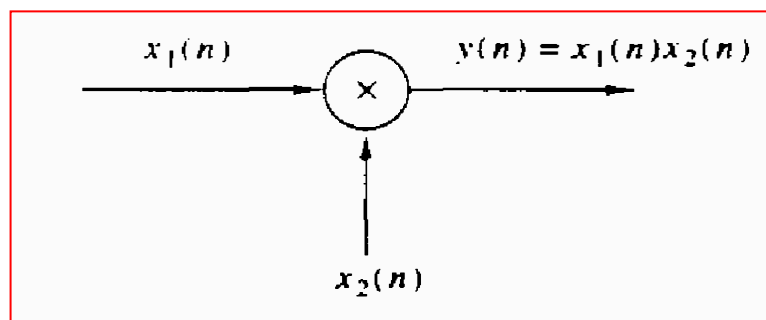


Note that it is not necessary to store either one of the sequences in order to perform the addition. In other words, the addition operation is memoryless.

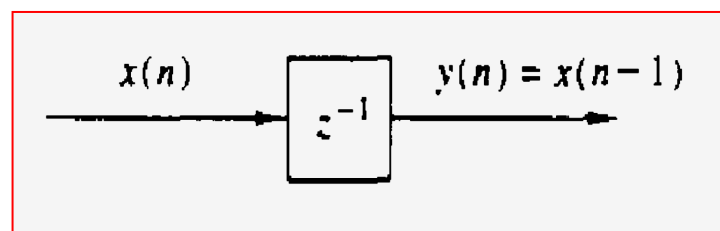
A constant multiplier. This operation is depicted by figure below, and simply represent applying a scale factor on the input $x(n)$. Note that this operation is also memoryless.



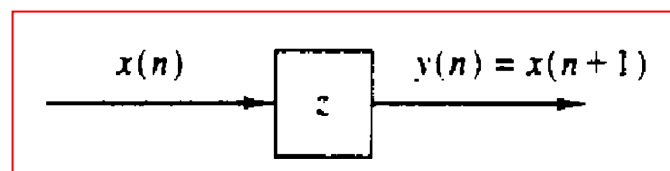
A signal multiplier. Figure below illustrates the multiplication of two signal sequences, the multiplication operation is memoryless.



A unit delay element. The unit delay is a special system that simply delays the signal passing through it by one sample



A unit advance element. In contrast to the unit delay, a unit advance moves the input a head by one sample.

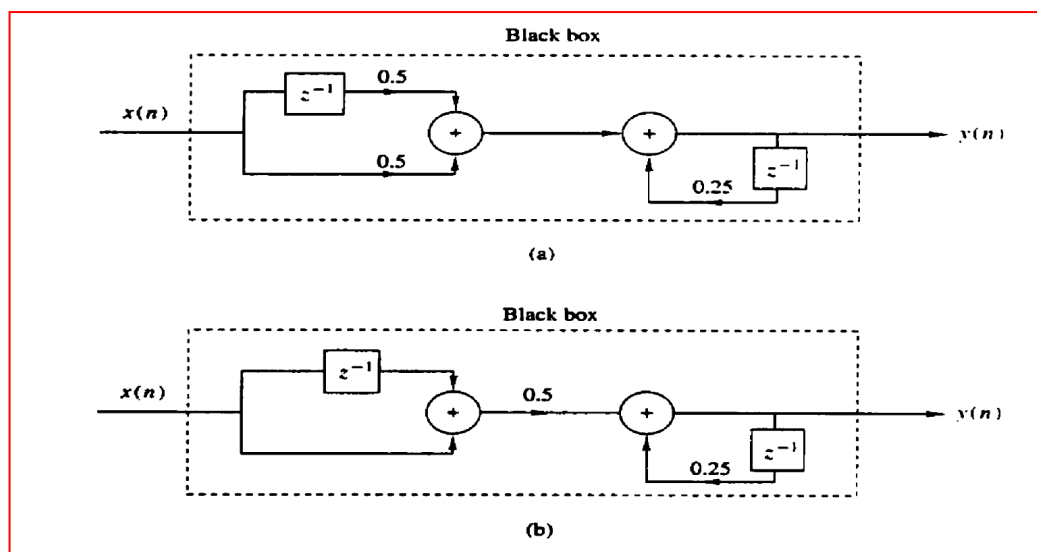


Example:-

Using basic building blocks introduced above, sketch the block diagram representation of the discrete – time system described by the input – output relation.

$$y(n] = \frac{1}{4}y[n - 1] + \frac{1}{2}x[n] + \frac{1}{2}x[n - 1]$$

Solution:-



Classification of Discrete – Time Systems:-

a) Static versus dynamic systems.

Static systems	Dynamic systems
$y(n) = ax(n)$ $y(n) = nx(n) + bx^3(n)$	$y(n) = x(n) + 3x(n - 1)$ $y(n) = \sum_{k=0}^n x(n - k)$

b) Time – Invariant versus Time – Variant systems

A relaxed system T is time invariant or shift invariant if and only if

$$x(n) \xrightarrow{T} y(n)$$

Implies that

$$x(n - k) \xrightarrow{T} y(n - k)$$

For every input signal $x(n)$ and every time shift k .

In general, we can write the output as

$$y(n, k) = T[x(n - k)]$$

Now if this output $y(n, k) = y(n - k)$, for all possible values of k , the system is time invariant. On the other hand, if the output $y(n, k) \neq y(n - k)$, even for one value of k , the system is time variant.

Example:-

The system described by the input – output equation

$$y(n) = T[x(n)] = x(n) - x(n - 1)$$

Now if the input is delayed by k units in time and applied to the system

$$y(n, k) = x(n - k) - x(n - k - 1)$$

On the other hand, if we delay $y(n)$ by k units in time, we obtain

$$y(n - k) = x(n - k) - x(n - k - 1)$$

Therefore, $y(n, k) = y(n - k)$ and the system is time invariant.

Example:-

The system described by the input – output equation

$$y(n) = T[x(n)] = x(-n)$$

The response of this system to $x(n - k)$ is

$$y(n, k) = T[x(n - k)] = x(-n - k)$$

Now, if we delay the output $y(n)$ by k units in time, the result will be

$$y(n - k) = x(-n + k)$$

Since $y(n, k) \neq y(n - k)$, the system is time variant.

c) Linear versus nonlinear systems.

A relaxed T system is linear if and only if

$$T[a_1x_1(n) + a_2x_2(n)] = a_1T[x_1(n)] + a_2T[x_2(n)]$$

For any arbitrary input sequences $x_1(n)$ and $x_2(n)$, and any arbitrary constants a_1 and a_2 .

Example:-

Determine if the systems described by the following input – output equations are linear or nonlinear.

a) $y(n) = nx(n)$

b) $y(n) = x^2(n)$

Solution:-

a) For two input sequences $x_1(n)$ and $x_2(n)$, the corresponding outputs are

$$y_1(n) = nx_1(n)$$

$$y_2(n) = nx_2(n)$$

A linear combination of the two input sequences results in the output

$$y_3(n) = T[a_1x_1(n) + a_2x_2(n)] = n[a_1x_1(n) + a_2x_2(n)]$$

A linear combination of the $y_1(n)$ and $y_2(n)$ results in

$$a_1y_1(n) + a_2y_2(n) = n[a_1x_1(n) + a_2x_2(n)]$$

Since $y_3(n) \equiv a_1y_1(n) + a_2y_2(n)$ the system is linear.

b) The responses of the system to two separate input signals are

$$y_1(n) = x_1^2(n)$$

$$y_2(n) = x_2^2(n)$$

The response of the system to a linear combination of these two input signals is

$$y_3(n) = T[a_1x_1(n) + a_2x_2(n)]$$

$$= [a_1x_1(n) + a_2x_2(n)]^2$$

$$= a_1^2x_1^2(n) + 2a_1a_2x_1(n)x_2(n) + a_2^2x_2^2(n)$$

On the other hand, if the system is linear, it would produce a linear combination of the two outputs

$$a_1 y_1(n) + a_2 y_2(n) = a_1 x_1^2(n) + a_2 x_2^2(n)$$

Since the actual output of the system, is not equal to the above equation, the system is nonlinear.

d) Causal versus noncausal systems

In mathematical terms, the output of a causal system satisfies an equation of the form

$$y(n) = F[x(n), x(n-1), x(n-2), \dots]$$

Where $F[\cdot]$ is some arbitrary function.

If a system does not satisfy this definition, it is called **noncausal**. Such a system has an output that depends not only on present and past inputs but also on future inputs.

Example:-

Determine if the systems described by the following input – output equations are causal or noncausal.

a) $y(n) = x(n) - x(n-1)$

b) $y(n) = \sum_{k=-\infty}^n x(k)$

c) $y(n) = ax(n)$

d) $y(n) = x(n) + 3x(n+4)$

d) $y(n) = x(n^2)$

f) $y(n) = x(2n)$

g) $y(n) = x(-n)$

Solution:-

The systems described by parts (a), (b), and (c) are causal.

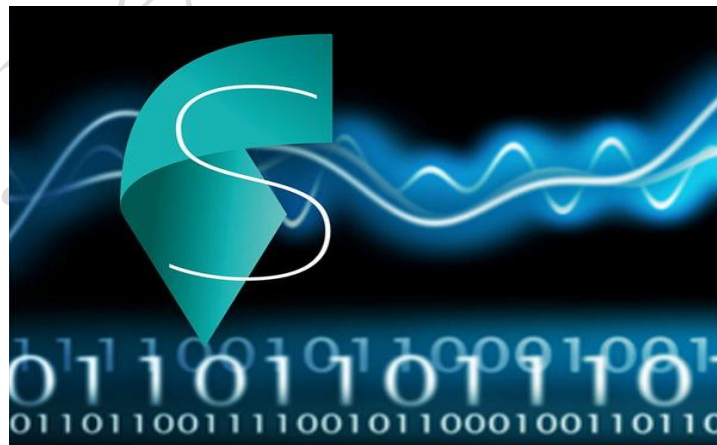
The systems described by rest parts are noncausal.



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Lect.6,7 Analysis of Discrete Linear Time-Invariant Systems.



Asst. Lect. Haider Saad

2025 - 2026

Difference Equations and Impulse Responses

Format of Difference Equation

A causal (The system is called causal because the current output depends only on the current input and past inputs and outputs, not on the future), linear time-invariant (The system is called Linear Time-Invariant (LTI) because it is linear and constant with time) system can be described by a difference equation having the following general form

$$\begin{aligned} y(n) + a_1y(n-1) + \dots + a_Ny(n-N) \\ = b_0x(n) + b_1x(n-1) + \dots + b_Mx(n-M), \end{aligned} \quad (1)$$

where $a_1, \dots, a_N$ and $b_0, b_1, \dots, b_M$ are the coefficients of the difference equation. Equation (1) can further be written as

$$\begin{aligned} y(n) = -a_1y(n-1) - \dots - a_Ny(n-N) \\ + b_0x(n) + b_1x(n-1) + \dots + b_Mx(n-M) \end{aligned} \quad (2)$$

Or

$$y(n) = -\sum_{i=1}^N a_iy(n-i) + \sum_{j=0}^M b_jx(n-j). \quad (3)$$

Notice that $y(n)$ is the current output, which depends on the past output samples $y(n-1), \dots, y(n-N)$, the current input sample $x(n)$, and the past input samples, $x(n-1), \dots, x(n-M)$.

We will examine the specific difference equations in the following examples.

Example

Given the following difference equation:

$$y(n) = 0.25y(n-1) + x(n),$$

Identify the nonzero system coefficients.

Solution:

Comparison with Equation (2) leads to

$$\begin{aligned} b_0 &= 1 \\ -a_1 &= 0.25, \end{aligned}$$

System Representation Using Its Impulse Response:

A linear time-invariant system can be completely described by its unit-impulse response, which is defined as the system response due to the impulse input $\delta(n)$ with zero initial conditions, depicted in Figure (1).

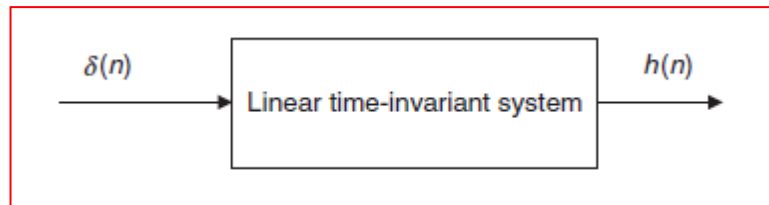


Figure (1): Unit-impulse response of the linear time-invariant system.

With the obtained unit-impulse response $h(n)$, we can represent the linear time-invariant system in Figure (2).

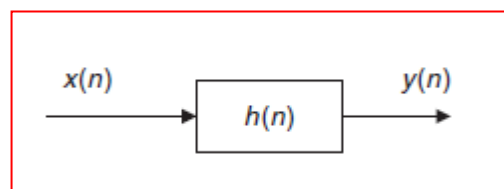
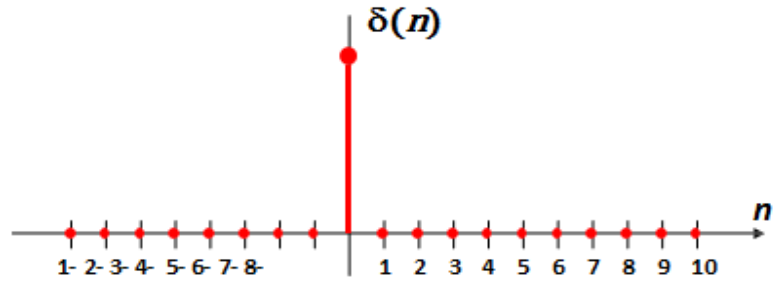


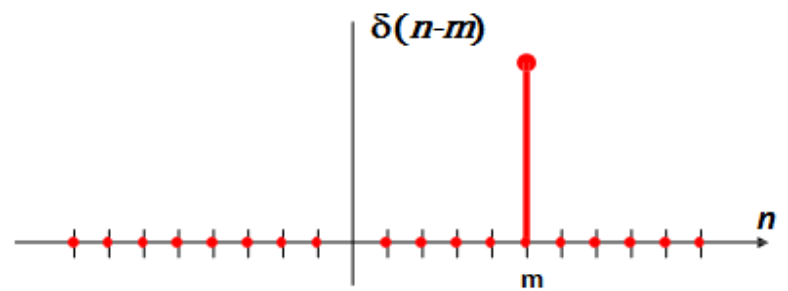
Figure (2): Representation of a linear time-invariant system using the impulse response.

❑ delta function or unit-impulse (sample) sequence $\delta(n)$

$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

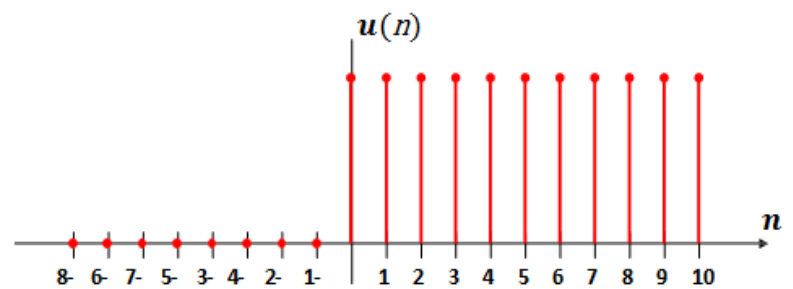


$$\delta(n-m) = \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases}$$

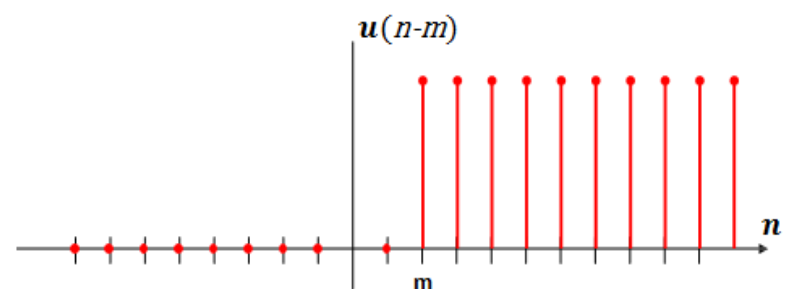


❑ unit-step sequence $U(n)$

$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



$$u(n-m) = \begin{cases} 1 & n \geq m \\ 0 & n < m \end{cases}$$



Example:

Given the linear time-invariant system

$$y(n) = 0.5x(n) + 0.25x(n - 1) \text{ with an initial condition } x(-1) = 0,$$

- Determine the unit-impulse response $h(n)$.
- Draw the system block diagram.
- Write the output using the obtained impulse response.

Solution:

- According to Figure 1, let $x(n) = \delta(n)$, then

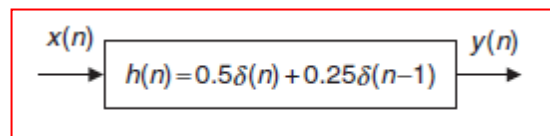
$$h(n) = y(n) = 0.5x(n) + 0.25x(n - 1) = 0.5\delta(n) + 0.25\delta(n - 1).$$

Thus, for this particular linear system, we have

$$h(n) = \begin{cases} 0.5 & n = 0 \\ 0.25 & n = 1 \\ 0 & \text{elsewhere} \end{cases}$$

note
unit response is $h(n)$
unit sequence is $\delta(n)$

- The block diagram of the linear time-invariant system is shown as



- The system output can be rewritten as

$$y(n) = h(0)x(n) + h(1)x(n - 1).$$

In general, we can express the output sequence of a linear time-invariant system from its impulse response and inputs as

$$y(n) = \dots + h(-1)x(n+1) + h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \dots \quad (3)$$

Equation (3) is called the **digital convolution sum**, which will be explored in a later section. We can verify Equation (3) by substituting the impulse sequence $x(n) = \delta(n)$ to get the impulse response

$$h(n) = \dots + h(-1)\delta(n+1) + h(0)\delta(n) + h(1)\delta(n-1) + h(2)\delta(n-2) + \dots,$$

Example:

Given the difference equation

$$y(n) = 0.25y(n-1) + x(n) \text{ for } n \geq 0 \text{ and } y(-1) = 0,$$

- Determine the unit-impulse response $h(n)$.
- Draw the system block diagram.
- Write the output using the obtained impulse response.
- For a step input $x(n) = u(n)$, verify and compare the output responses for the first three output samples using the difference equation and digital convolution sum (Equation 3).

Solution:

- Let $x(n) = \delta(n)$, then

$$h(n) = 0.25h(n-1) + \delta(n).$$

To solve for $h(n)$, we evaluate

$$h(0) = 0.25h(-1) + \delta(0) = 0.25 \times 0 + 1 = 1$$

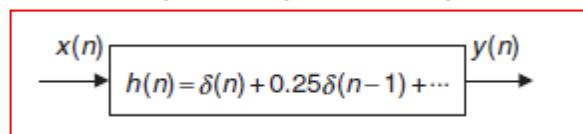
$$h(1) = 0.25h(0) + \delta(1) = 0.25 \times 1 + 0 = 0.25$$

$$h(2) = 0.25h(1) + \delta(2) = 0.25 \times 0.25 + 0 = 0.0625$$

With the calculated results, we can predict the impulse response as

$$h(n) = \delta(n) + 0.25\delta(n-1) + 0.0625\delta(n-2) + \dots$$

- The system block diagram is given in Figure below.



c) The output sequence is a sum of infinite terms expressed as

$$\begin{aligned} y(n) &= h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \dots \\ &= x(n) + 0.25x(n-1) + 0.0625x(n-2) + \dots \end{aligned}$$

d) From the difference equation and using the zero-initial condition, we have

$$\begin{aligned} y(n) &= 0.25y(n-1) + x(n) \text{ for } n \geq 0 \text{ and } y(-1) = 0 \\ n = 0, y(0) &= 0.25y(-1) + x(0) = u(0) = 1 \\ n = 1, y(1) &= 0.25y(0) + x(1) = 0.25 \times u(0) + u(1) = 1.25 \\ n = 2, y(2) &= 0.25y(1) + x(2) = 0.25 \times 1.25 + u(2) = 1.3125 \\ &\dots \end{aligned}$$

Applying the convolution sum in Equation (3) yields

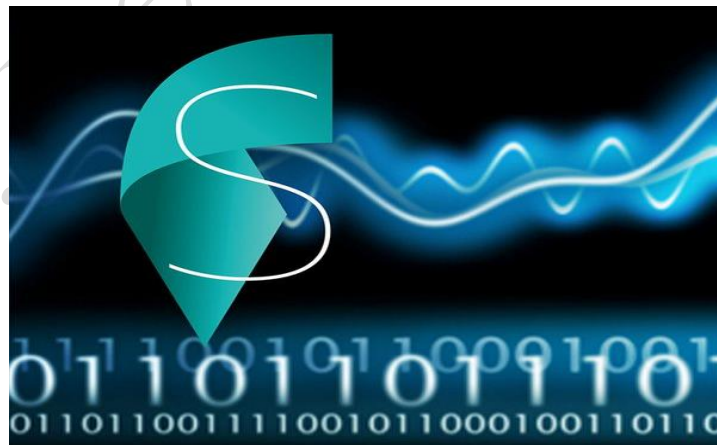
$$\begin{aligned} y(n) &= x(n) + 0.25x(n-1) + 0.0625x(n-2) + \dots \\ n = 0, y(0) &= x(0) + 0.25x(-1) + 0.0625x(-2) + \dots \\ &= u(0) + 0.25 \times u(-1) + 0.125 \times u(-2) + \dots = 1 \\ n = 1, y(1) &= x(1) + 0.25x(0) + 0.0625x(-1) + \dots \\ &= u(1) + 0.25 \times u(0) + 0.125 \times u(-1) + \dots = 1.25 \\ n = 2, y(2) &= x(2) + 0.25x(1) + 0.0625x(0) + \dots \\ &= u(2) + 0.25 \times u(1) + 0.0625 \times u(0) + \dots = 1.3125 \end{aligned}$$



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DIGITAL SIGNAL PROCESSING THIRD STAGE

Lect.8 Digital Convolution and Deconvolution.



Asst. Lect. Haider Saad

2025 - 2026

Digital Convolution

Given a linear time-invariant system, we can determine its unit-impulse response $h(n)$, which relates the system input and output. To find the output sequence $y(n)$ for any input sequence $x(n)$, we write the digital convolution as shown in Equation (3) as:

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) \\ &= \dots + h(-1)x(n+1) + h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \dots \\ y(n) &= \sum_{k=-\infty}^{\infty} x(k)h(n-k) \\ &= \dots + x(-1)h(n+1) + x(0)h(n) + x(1)h(n-1) + x(2)h(n-2) + \dots \end{aligned} \tag{4}$$

Using a conventional notation, we express the digital convolution as

$$y(n) = h(n) * x(n). \tag{5}$$

Note that for a causal system, which implies its impulse response

$$h(n) = 0 \text{ for } n < 0,$$

The lower limit of the convolution sum begins at 0 instead of 1, that is

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k) = \sum_{k=0}^{\infty} x(k)h(n-k).$$

We will focus on evaluating the convolution sum based on Equation (4). Let us examine first a few outputs from Equation (4):

$$\begin{aligned} y(0) &= \sum_{k=-\infty}^{\infty} x(k)h(-k) = \dots + x(-1)h(1) + x(0)h(0) + x(1)h(-1) + x(2)h(-2) + \dots \\ y(1) &= \sum_{k=-\infty}^{\infty} x(k)h(1-k) = \dots + x(-1)h(2) + x(0)h(1) + x(1)h(0) + x(2)h(-1) + \dots \\ y(2) &= \sum_{k=-\infty}^{\infty} x(k)h(2-k) = \dots + x(-1)h(3) + x(0)h(2) + x(1)h(1) + x(2)h(0) + \dots \\ &\dots \end{aligned}$$

Digital convolution using the graphical method.

Step 1. Obtain the reversed sequence $h(-k)$.

Step 2. Shift $h(-k)$ by $|n|$ samples to get $h(n-k)$. If $n \geq 0$, $h(-k)$ will be shifted to the right by n samples; but if $n < 0$, $h(-k)$ will be shifted to the left by $|n|$ samples.

Step 3. Perform the convolution sum that is the sum of the products of two sequences $x(k)$ and $h(n-k)$ to get $y(n)$.

Step 4. Repeat steps 1 to 3 for the next convolution value $y(n)$.

Example

Given a sequence,

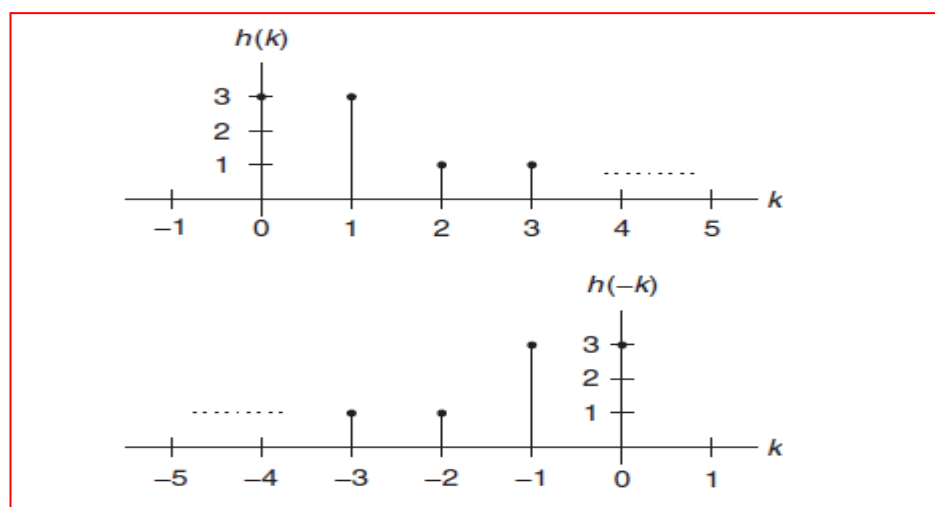
$$h(k) = \begin{cases} 3, & k = 0, 1 \\ 1, & k = 2, 3 \\ 0 & \text{elsewhere} \end{cases}$$

where k is the time index or sample number,

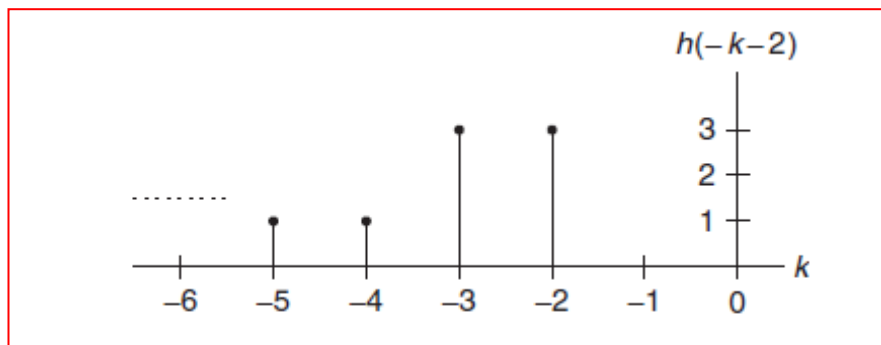
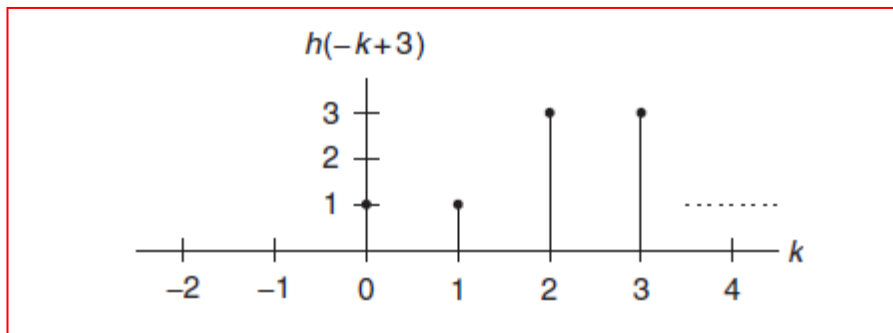
- a) Sketch the sequence $h(k)$ and reversed sequence $h(-k)$.
- b) Sketch the shifted sequences $h(-k+3)$ and $h(-k-2)$.

Solution:

a)

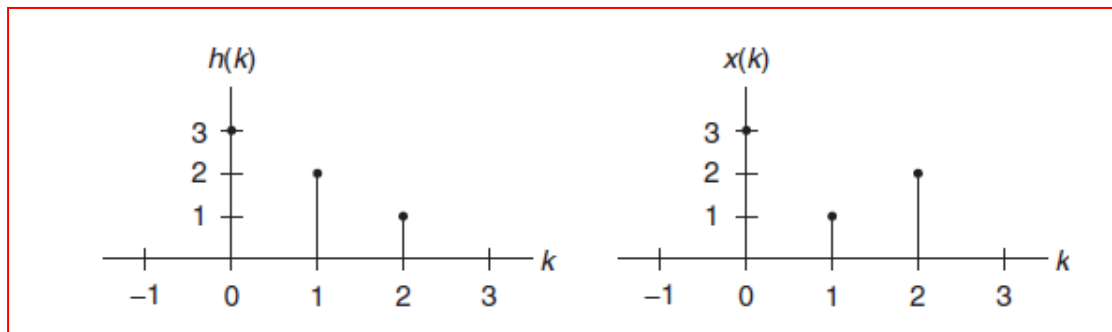


b)



Example

Using the following sequences defined in the Figure below, evaluate the digital convolution



$$h(k) = \begin{bmatrix} 3 & 2 & 1 \end{bmatrix}$$

$\uparrow$
 0

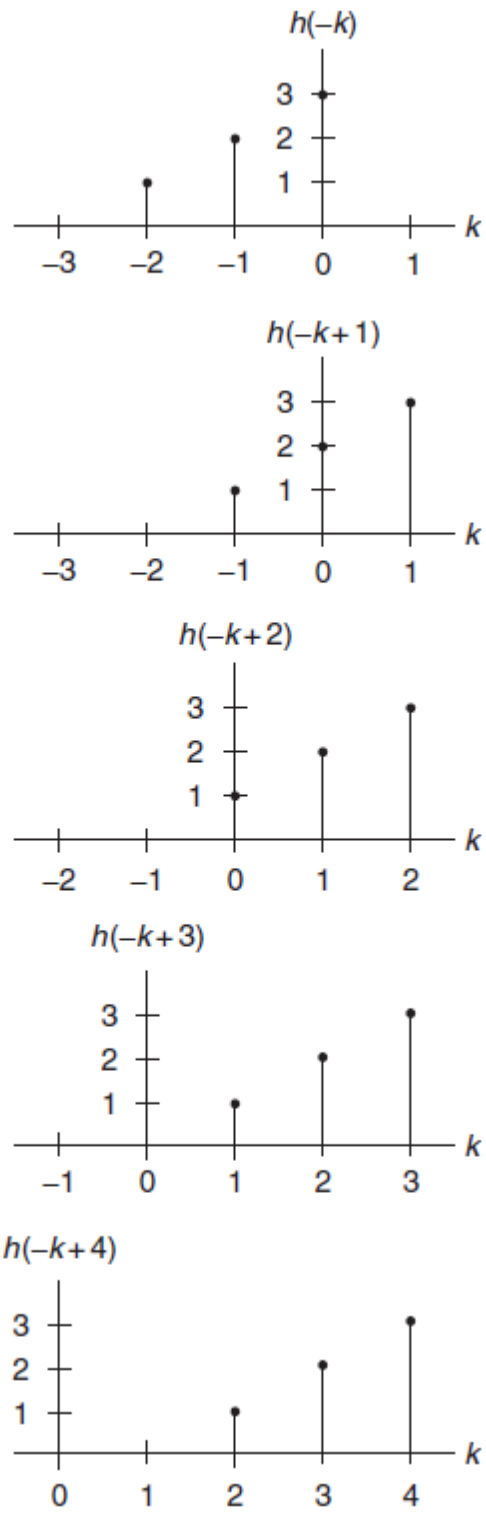
$$x(k) = \begin{bmatrix} 3 & 1 & 2 \end{bmatrix}$$

$\uparrow$
 0

- a) By the graphical method.
- b) By applying the formula directly.

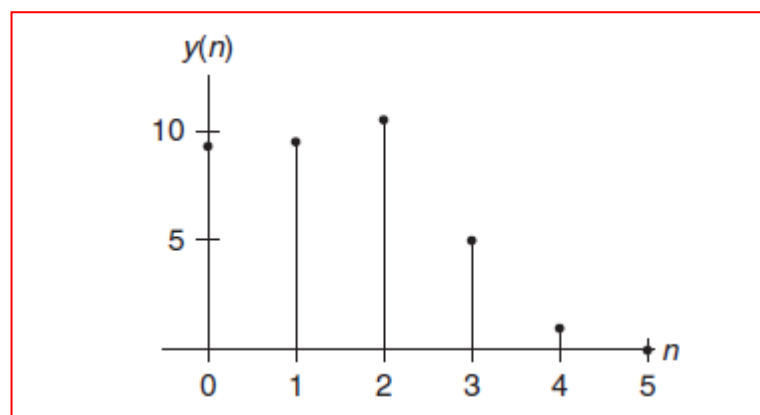
Solution:

a)



We can compute the convolution sum as:

sum of product of $x(k)$ and $h(-k)$: $y(0) = 3 \times 3 = 9$
 sum of product of $x(k)$ and $h(1-k)$: $y(1) = 1 \times 3 + 3 \times 2 = 9$
 sum of product of $x(k)$ and $h(2-k)$: $y(2) = 2 \times 3 + 1 \times 2 + 3 \times 1 = 11$
 sum of product of $x(k)$ and $h(3-k)$: $y(3) = 2 \times 2 + 1 \times 1 = 5$
 sum of product of $x(k)$ and $h(4-k)$: $y(4) = 2 \times 1 = 2$
 sum of product of $x(k)$ and $h(5-k)$: $y(n) = 0$ for $n > 4$, since sequences $x(k)$ and $h(n-k)$ do not overlap.



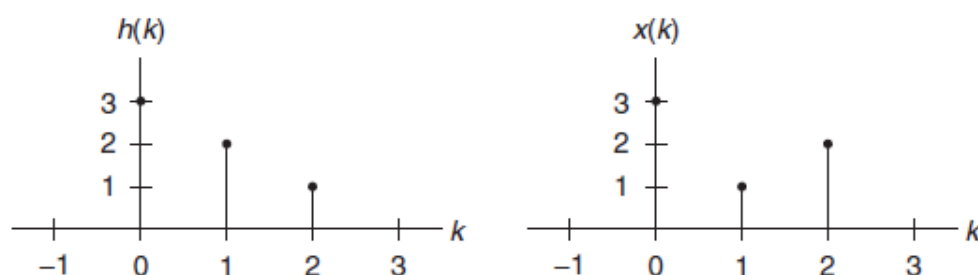
b) Applying Equation (4) with zero initial conditions leads to

$$y(n) = x(0)h(n) + x(1)h(n-1) + x(2)h(n-2)$$

$n = 0$, $y(0) = x(0)h(0) + x(1)h(-1) + x(2)h(-2) = 3 \times 3 + 1 \times 0 + 2 \times 0 = 9$,
 $n = 1$, $y(1) = x(0)h(1) + x(1)h(0) + x(2)h(-1) = 3 \times 2 + 1 \times 3 + 2 \times 0 = 9$,
 $n = 2$, $y(2) = x(0)h(2) + x(1)h(1) + x(2)h(0) = 3 \times 1 + 1 \times 2 + 2 \times 3 = 11$,
 $n = 3$, $y(3) = x(0)h(3) + x(1)h(2) + x(2)h(1) = 3 \times 0 + 1 \times 1 + 2 \times 2 = 5$.
 $n = 4$, $y(4) = x(0)h(4) + x(1)h(3) + x(2)h(2) = 3 \times 0 + 1 \times 0 + 2 \times 1 = 2$,
 $n \geq 5$, $y(n) = x(0)h(n) + x(1)h(n-1) + x(2)h(n-2) = 3 \times 0 + 1 \times 0 + 2 \times 0 = 0$.

Example:

Using the following sequences defined in the Figure below, evaluate the digital convolution



Convolve them using the table method.

Solution:

Digital convolution steps via the table.

- Step 1. List the index k covering a sufficient range.
- Step 2. List the input $x(k)$.
- Step 3. Obtain the reversed sequence $h(-k)$, and align the rightmost element of $h(n-k)$ to the leftmost element of $x(k)$.
- Step 4. Cross-multiply and sum the nonzero overlap terms to produce $y(n)$.
- Step 5. Slide $h(n-k)$ to the right by one position.
- Step 6. Repeat step 4; stop if all the output values are zero or if required.

Convolution sum using the table method.

k :	-2	-1	0	1	2	3	4	5	
$x(k)$:			3	1	2				
$h(-k)$:	1	2	3						$y(0) = 3 \times 3 = 9$
$h(1-k)$:		1	2	3					$y(1) = 3 \times 2 + 1 \times 3 = 9$
$h(2-k)$:			1	2	3				$y(2) = 3 \times 1 + 1 \times 2 + 2 \times 3 = 11$
$h(3-k)$:				1	2	3			$y(3) = 1 \times 1 + 2 \times 2 = 5$
$h(4-k)$:					1	2	3		$y(4) = 2 \times 1 = 2$
$h(5-k)$:						1	2	3	$y(5) = 0$ (no overlap)

Deconvolution:

The digital Deconvolution can be performed by *Iterative Approach*, *Polynomial Approach*, and *Graphical Method*. In the following subsection, the polynomial approach will be explained.

Polynomial Approach:

A long division process is applied between two polynomials. For causal system, the remainder is always zero.

Example: If $y(n) = [15 \ -8 \ -5 \ 2]$ and $h(n) = [-3 \ 1]$ find $x(n)$.

Solution: $y = 15 - 8x - 5x^2 + 2x^3$, and $h = -3 + x$. Applying long division, we obtain

$$\begin{array}{r} 2x^2 + x - 5 \\ x-3 \overline{) 2x^3 - 5x^2 - 8x + 15} \\ \underline{2x^3 - 6x^2} \\ x^2 - 8x \\ \underline{x^2 - 3x} \\ -5x + 15 \\ \underline{-5x + 15} \\ \text{Remainder } 0 \end{array}$$

result $= -5 + x + 2x^2$. Then $x(n) = [-5 \ 1 \ 2]$

Example: If $y(n) = [12 \ 10 \ 14 \ 6]$ and $h(n) = [4 \ 2]$ find $x(n)$.

Solution: $y = 12 + 10x + 14x^2 + 6x^3$, and $h = 4 + 2x$. Applying long division, we obtain

result $= 3 + x + 3x^2$. Then $x(n) = [3 \ 1 \ 3]$

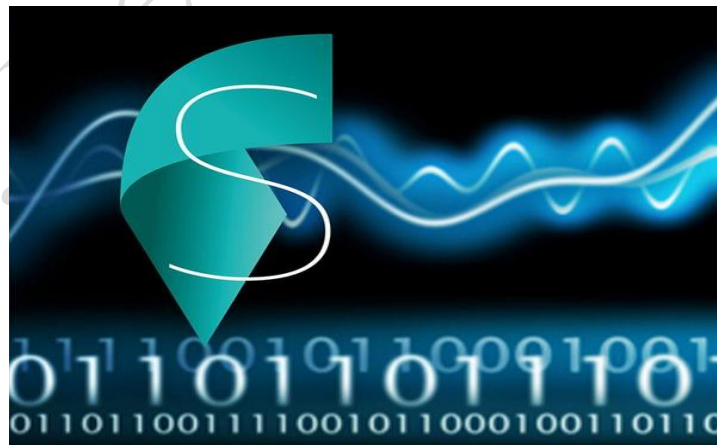


CYBER SECURITY TECHNOLOGY ENGINEERING DEPARTMENT

DIGITAL SIGNAL PROCESSING

Third Stage

Lect.9 Discrete Fourier Transform

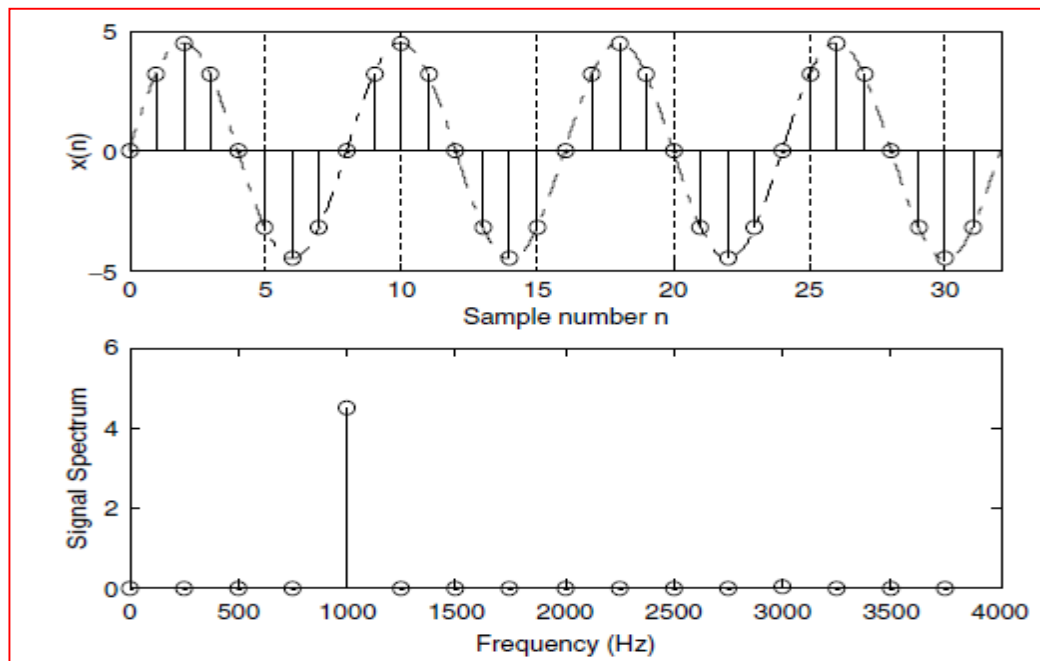


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Discrete Fourier Transform

Figure below illustrates the time domain representation of a 1,000-Hz sinusoid with 32 samples at a sampling rate of 8,000 Hz; the bottom plot shows the signal spectrum (frequency domain representation), where we can clearly observe that the amplitude peak is located at the frequency of 1,000 Hz in the calculated spectrum. Hence, the spectral plot better displays frequency information of a digital signal.

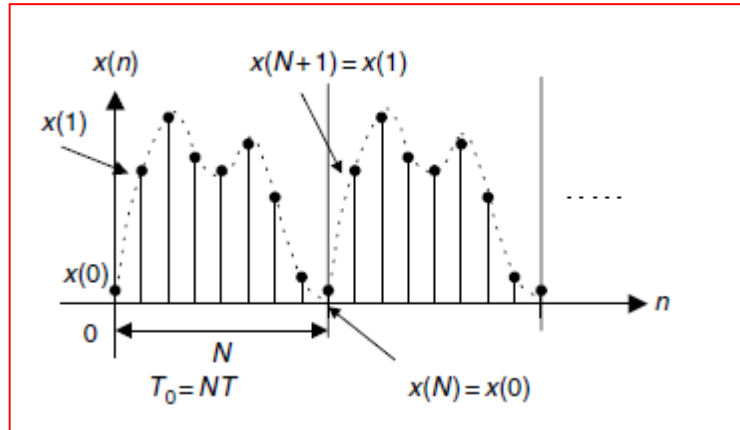


The algorithm transforming the time domain signal samples to the frequency domain components is known as the discrete Fourier transform, or DFT.

The DFT also establishes a relationship between the time domain representation and the frequency domain representation. Therefore, we can apply the DFT to perform frequency analysis of a time domain sequence.

Fourier Series Coefficients of Periodic Digital Signals

Let us look at a process in which we want to estimate the spectrum of a periodic digital signal $x(n)$ sampled at a rate of f_s Hz with the fundamental period $T_0 = NT$, as shown in Figure below.



According to Fourier series analysis, the coefficients of the Fourier series expansion of a periodic signal $x(t)$ in a complex form is

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt \quad -\infty < k < \infty,$$

where k is the number of harmonics corresponding to the harmonic frequency of kf_0 and $\omega_0 = 2\pi / T_0$ and $f_0 = 1/T_0$ are the fundamental frequency in radians per second and the fundamental frequency in Hz, respectively.

To apply above Equation, we substitute $T_0 = NT$, $\omega_0 = 2\pi/T_0$ and approximate the integration over one period using a summation by substituting $dt = T$ and $t = nT$. We obtain

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}}, \quad -\infty < k < \infty.$$

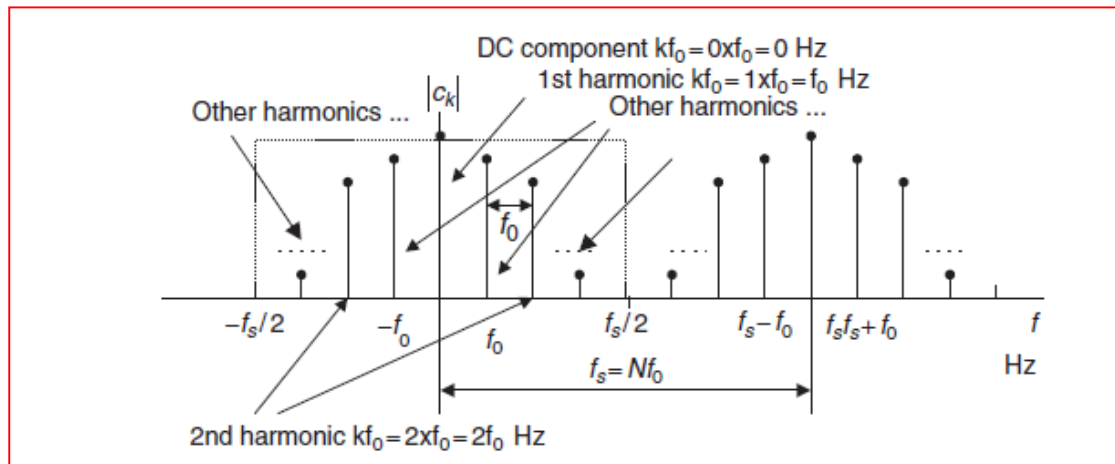
Since the coefficients c_k are obtained from the Fourier series expansion in the complex form, the resultant spectrum c_k will have two sides. There is an important feature of above Equation in which the Fourier series coefficient c_k is periodic of N . We can verify this as follows

$$c_{k+N} = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi(k+N)n}{N}} = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi k n}{N}} e^{-j 2\pi n}.$$

Since $e^{-j 2\pi n} = \cos(2\pi n) - j \sin(2\pi n) = 1$, it follows that

$$c_{k+N} = c_k.$$

Therefore, the two-sided line amplitude spectrum $|c_k|$ is periodic, as shown in Figure below.



For convenience, we compute the spectrum over the range from 0 to f_s Hz with nonnegative indices, that is,

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi k n}{N}}, \quad k = 0, 1, \dots, N-1.$$

Example

The periodic signal

$$x(t) = \sin(2\pi t)$$

is sampled using the rate $f_s = 4$ Hz.

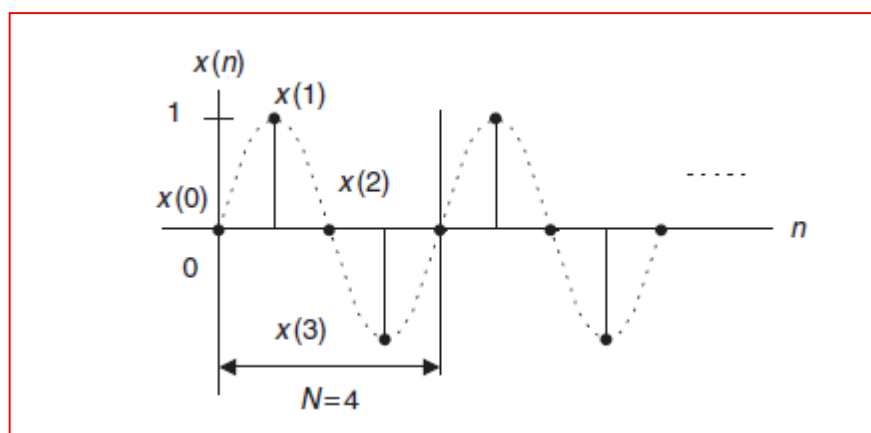
- Compute the spectrum c_k using the samples in one period.
- Plot the two-sided amplitude spectrum $|c_k|$ over the range from -2 to 2 Hz.

Solution:

- From the analog signal, we can determine the fundamental frequency $\omega_0 = 2\pi$ radians per second and $f_0 = \frac{\omega_0}{2\pi} = \frac{2\pi}{2\pi} = 1\text{Hz}$, and the fundamental period $T_0 = 1$ second. Since using the sampling interval $T = 1/f_s = 0.25$ second, we get the sampled signal as

$$x(n) = x(nT) = \sin(2\pi nT) = \sin(0.5\pi n)$$

and plot the first eight samples as shown in Figure



Choosing the duration of one period, $N = 4$, we have the sample values as follows

$$x(0) = 0; x(1) = 1; x(2) = 0; \text{ and } x(3) = -1.$$

$$c_0 = \frac{1}{4} \sum_{n=0}^3 x(n) = \frac{1}{4} (x(0) + x(1) + x(2) + x(3)) = \frac{1}{4} (0 + 1 + 0 - 1) = 0$$

$$c_1 = \frac{1}{4} \sum_{n=0}^3 x(n) e^{-j2\pi \times 1n/4} = \frac{1}{4} (x(0) + x(1)e^{-j\pi/2} + x(2)e^{-j\pi} + x(3)e^{-j3\pi/2})$$

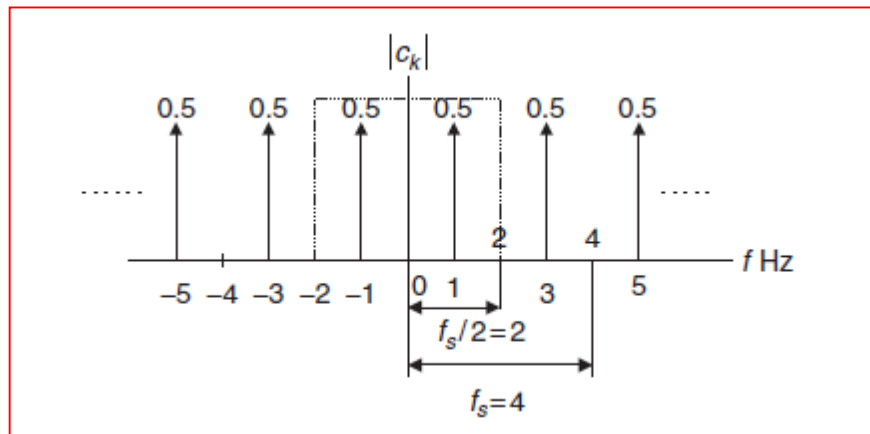
$$= \frac{1}{4} (x(0) - jx(1) - x(2) + jx(3)) = 0 - j(1) - 0 + j(-1) = -j0.5.$$

Similarly, we get

$$c_2 = \frac{1}{4} \sum_{k=0}^3 x(n) e^{-j2\pi \times 2n/4} = 0, \text{ and } c_3 = \frac{1}{4} \sum_{n=0}^3 x(k) e^{-j2\pi \times 3n/4} = j0.5.$$

Using periodicity, it follows that

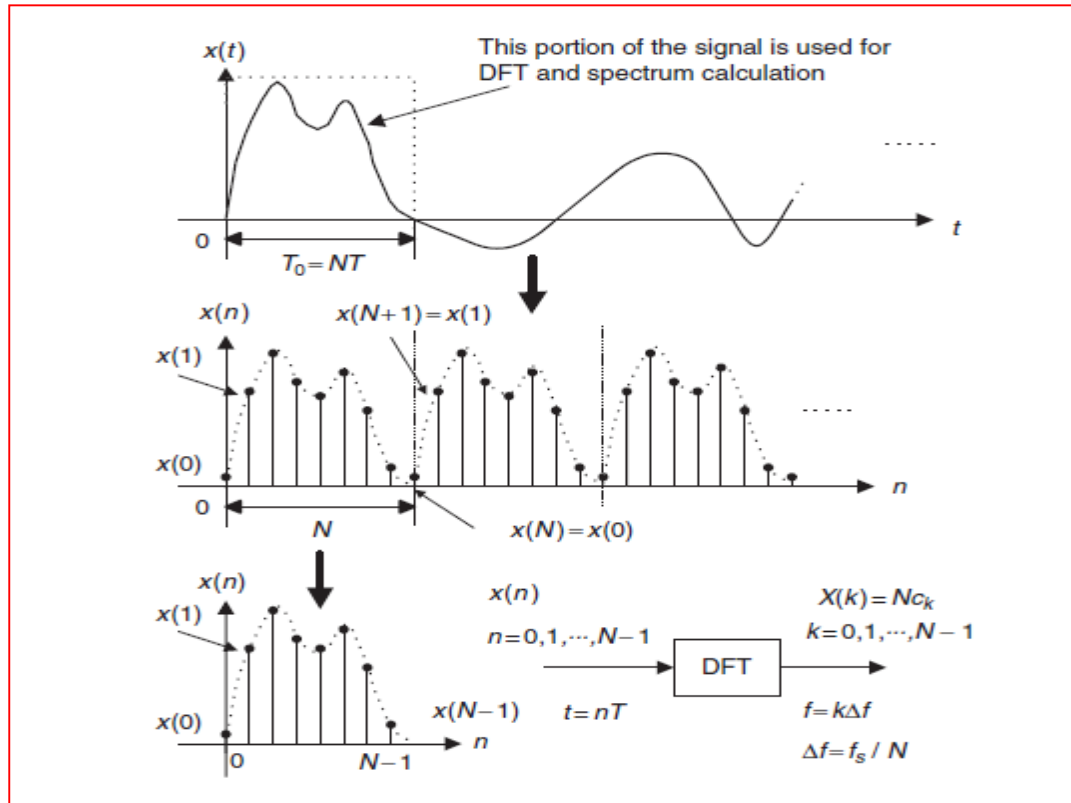
- b. The amplitude spectrum for the digital signal is sketched in Figure



As we know, the spectrum in the range of -2 to 2 Hz presents the information of the sinusoid with a frequency of 1 Hz and a peak value of $2|c_1| = 1$ which is converted from two sides to one side by doubling the spectral value. Note that we do not double the direct-current (DC) component, that is, c_0 .

Discrete Fourier Transform Formulas

Now, let us concentrate on development of the DFT. Figure below shows one way to obtain the DFT formula.



First, we assume that the process acquires data samples from digitizing the interested continuous signal for duration of T seconds. Next, we assume that a periodic signal $x(n)$ is obtained by copying the acquired N data samples with the duration of T to itself repetitively.

We determine the Fourier series coefficients using one-period N data samples then we multiply the Fourier series coefficients by a factor of N to obtain

$$X(k) = Nc_k = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}}, \quad k = 0, 1, \dots, N-1,$$

Now let us conclude the DFT definition. Given a sequence $x(n)$, $0 \leq n \leq N-1$, its DFT is defined as

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} = \sum_{n=0}^{N-1} x(n)W_N^{kn}, \text{ for } k = 0, 1, \dots, N-1.$$

Above equation can be expanded as

$$X(k) = x(0)W_N^{k0} + x(1)W_N^{k1} + x(2)W_N^{k2} + \dots + x(N-1)W_N^{k(N-1)},$$

for $k = 0, 1, \dots, N-1$,

Where the factor W_N (called the twiddle factor in some textbooks) is defined as

$$W_N = e^{-j2\pi/N} = \cos\left(\frac{2\pi}{N}\right) - j\sin\left(\frac{2\pi}{N}\right).$$

The inverse DFT is given by

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{j2\pi kn/N} = \frac{1}{N} \sum_{k=0}^{N-1} X(k)W_N^{-kn}, \text{ for } n = 0, 1, \dots, N-1.$$

Similarly

$$x(n) = \frac{1}{N} \left(X(0)W_N^{-0n} + X(1)W_N^{-1n} + X(2)W_N^{-2n} + \dots + X(N-1)W_N^{-(N-1)n} \right),$$

for $n = 0, 1, \dots, N-1$.

Example

Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 1$, $x(1) = 2$, $x(2) = 3$, and $x(3) = 4$,

Evaluate its DFT $X(k)$.

Solution:

$$X(k) = \sum_{n=0}^3 x(n) W_4^{kn} = \sum_{n=0}^3 x(n) e^{-j\frac{\pi kn}{2}}.$$

Thus, for $k = 0$

$$\begin{aligned} X(0) &= \sum_{n=0}^3 x(n) e^{-j0} = x(0) e^{-j0} + x(1) e^{-j0} + x(2) e^{-j0} + x(3) e^{-j0} \\ &= x(0) + x(1) + x(2) + x(3) \\ &= 1 + 2 + 3 + 4 = 10 \end{aligned}$$

for $k = 1$

$$\begin{aligned} X(1) &= \sum_{n=0}^3 x(n) e^{-j\frac{\pi n}{2}} = x(0) e^{-j0} + x(1) e^{-j\frac{\pi}{2}} + x(2) e^{-j\pi} + x(3) e^{-j\frac{3\pi}{2}} \\ &= x(0) - jx(1) - x(2) + jx(3) \\ &= 1 - j2 - 3 + j4 = -2 + j2 \end{aligned}$$

for $k = 2$

$$\begin{aligned} X(2) &= \sum_{n=0}^3 x(n) e^{-j\pi n} = x(0) e^{-j0} + x(1) e^{-j\pi} + x(2) e^{-j2\pi} + x(3) e^{-j3\pi} \\ &= x(0) - x(1) + x(2) - x(3) \\ &= 1 - 2 + 3 - 4 = -2 \end{aligned}$$

and for $k = 3$

$$\begin{aligned} X(3) &= \sum_{n=0}^3 x(n) e^{-j\frac{3\pi n}{2}} = x(0) e^{-j0} + x(1) e^{-j\frac{3\pi}{2}} + x(2) e^{-j3\pi} + x(3) e^{-j\frac{9\pi}{2}} \\ &= x(0) + jx(1) - x(2) - jx(3) \\ &= 1 + j2 - 3 - j4 = -2 - j2 \end{aligned}$$

Example

Using the DFT coefficients $X(k)$ for $0 \leq k \leq 3$ computed in previous example

Evaluate its inverse DFT to determine the time domain sequence $x(n)$.

Solution:

$$x(n) = \frac{1}{4} \sum_{k=0}^3 X(k) W_4^{-nk} = \frac{1}{4} \sum_{k=0}^3 X(k) e^{j\frac{\pi kn}{2}}.$$

Then for $n = 0$

$$\begin{aligned} x(0) &= \frac{1}{4} \sum_{k=0}^3 X(k) e^{j0} = \frac{1}{4} (X(0)e^{j0} + X(1)e^{j0} + X(2)e^{j0} + X(3)e^{j0}) \\ &= \frac{1}{4} (10 + (-2 + j2) - 2 + (-2 - j2)) = 1 \end{aligned}$$

for $n = 1$

$$\begin{aligned} x(1) &= \frac{1}{4} \sum_{k=0}^3 X(k) e^{j\frac{k\pi}{2}} = \frac{1}{4} (X(0)e^{j0} + X(1)e^{j\frac{\pi}{2}} + X(2)e^{j\pi} + X(3)e^{j\frac{3\pi}{2}}) \\ &= \frac{1}{4} (X(0) + jX(1) - X(2) - jX(3)) \\ &= \frac{1}{4} (10 + j(-2 + j2) - (-2) - j(-2 - j2)) = 2 \end{aligned}$$

for $n = 2$

$$\begin{aligned} x(2) &= \frac{1}{4} \sum_{k=0}^3 X(k) e^{jk\pi} = \frac{1}{4} (X(0)e^{j0} + X(1)e^{j\pi} + X(2)e^{j2\pi} + X(3)e^{j3\pi}) \\ &= \frac{1}{4} (X(0) - X(1) + X(2) - X(3)) \\ &= \frac{1}{4} (10 - (-2 + j2) + (-2) - (-2 - j2)) = 3 \end{aligned}$$

and for $n = 3$

$$\begin{aligned}x(3) &= \frac{1}{4} \sum_{k=0}^3 X(k) e^{j \frac{k\pi 3}{2}} = \frac{1}{4} \left(X(0) e^{j0} + X(1) e^{j \frac{3\pi}{2}} + X(2) e^{j3\pi} + X(3) e^{j \frac{9\pi}{2}} \right) \\&= \frac{1}{4} (X(0) - jX(1) - X(2) + jX(3)) \\&= \frac{1}{4} (10 - j(-2 + j2) - (-2) + j(-2 - j2)) = 4\end{aligned}$$

Now we explore the relationship between the frequency bin k and its associated frequency.

The calculated N DFT coefficients $X(k)$ represent the frequency components ranging from 0 Hz (or radians/second) to f_s Hz (or ω_s radians/second), hence we can map the frequency bin k to its corresponding frequency as follows:

$$\omega = \frac{k\omega_s}{N} \text{ (radians per second),}$$

or in terms of Hz,

$$f = \frac{kf_s}{N} \text{ (Hz),}$$

where $\omega_s = 2\pi f_s$.

We can define the frequency resolution as the frequency step between two consecutive DFT coefficients to measure how fine the frequency domain presentation is and achieve

$$\Delta\omega = \frac{\omega_s}{N} \text{ (radians per second),}$$

or in terms of Hz, it follows that

$$\Delta f = \frac{f_s}{N} \text{ (Hz).}$$

Example

In previous examples, given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 1$, $x(1) = 2$, $x(2) = 3$, and $x(3) = 4$, we have computed four DFT coefficients $X(k)$ for $0 \leq k \leq 3$ as $X(0) = 10$, $X(1) = -2 + j2$, $X(2) = -2$, and $X(3) = -2 - j2$. If the sampling rate is 10 Hz,

- a. Determine the sampling period, time index, and sampling time instant for a digital sample $x(3)$ in time domain.
- b. Determine the frequency resolution, frequency bin number, and mapped frequency for each of the DFT coefficients $X(1)$ and $X(3)$ in frequency domain.

Solution:

- a. In time domain, we have the sampling period calculated as

$$T = 1/f_s = 1/10 = 0.1 \text{ second.}$$

For data $x(3)$, the time index is $n = 3$ and the sampling time instant is determined by

$$t = nT = 3 \cdot 0.1 = 0.3 \text{ second.}$$

- b. In frequency domain, since the total number of DFT coefficients is four, the frequency resolution is determined by

$$\Delta f = \frac{f_s}{N} = \frac{10}{4} = 2.5 \text{ Hz.}$$

The frequency bin number for $X(1)$ should be $k = 1$ and its corresponding frequency is determined by

$$f = \frac{kf_s}{N} = \frac{1 \times 10}{4} = 2.5 \text{ Hz.}$$

Similarly, for $X(3)$ and $k = 3$,

$$f = \frac{kf_s}{N} = \frac{3 \times 10}{4} = 7.5 \text{ Hz.}$$